Department of Mathematics
TUM School of Computation, Information and Technology
Technical University of Munich

Topological indices of Quantum Walks

Topologische Indizes von Quantum Walks

Andreas Schäfer

Thesis for the attainment of the academic degree

Master of Science

at the TUM School of Computation, Information and Technology of the Technical University of Munich

Supervisor:
Prof. Dr. Simone Warzel

Advisors:
Prof. Dr. Simone Warzel

Submitted:
Munich, 1. August 2023

I hereby declare that this thesis is entirely the result of my own work except where otherwise indicated. I have only used the resources given in the list of references.

Munich, 1. August 2023 Andreas Schäfer

Zusammenfassung

Die Quantum Walks, mit denen wir uns beschäftigen, sind lokale, translationsinvariante, unitäre Operatoren U auf dem Hilbert Raum $l^2(\mathbb{Z}^d;\mathbb{C}^s)$. Ihre Fourier Transformation ist ein Multiplikationsoperator mit der Matrix $\hat{U}(k)$, deren Determinante ein Monom der Form $C\,e^{iv\cdot k}$ ist. Den ganzzahligen Vektor $v \in \mathbb{Z}^d$ nennen wir den Index von U. Er wurde zuerst von Kitaev als Fluss einer Unitären eingeführt [Kit06]. Der Index entspricht dem durchschnittlichen Informationsfluss und ist eine Homotopie Invariante. Zusätzlich lässt er sich über die Bänder von $\hat{U}(k)$ beschreiben. Im Eindimensionalen (d.h. $d = 1$) ist der Index gegeben durch die Spur des Operators $U^\dagger PU - P$. Hier steht P für die Projektion auf $\mathbb{Z}_{\geq 0}$. Mittels der Indextheorie eines Paares von Projektionen [ASS94] kann man nun den Index auch für nicht translationsinvariante Operatoren definieren. Im Mehrdimensionalen (d.h. $d > 1$) gilt eine schwächere Variante dieses Resultats: Hier lässt sich der Index von U mittels einer Projektion P_L bestimmen, die von der Reichweite L von U abhängt.

Abstract

The Quantum Walks we will study are local, translation invariant, unitary operators U on $l^2(\mathbb{Z}^d;\mathbb{C}^s)$. Their Fourier transform is a multiplication operator by a matrix $\hat{U}(k)$, whose determinant is a monomial $C\,e^{iv\cdot k}$. This integer vector $v \in \mathbb{Z}^d$ will be called the index of a Quantum Walk and was first introduced by Kitaev as the flow of a unitary [Kit06]. It corresponds to the average flow of information and is a homotopy invariant. Furthermore, we can derive it using the bands of $\hat{U}(k)$. In the one-dimensional case (i.e. $d = 1$), the index is given by the trace of the operator $U^\dagger PU - P$, where P is the projection onto $\mathbb{Z}_{\geq 0}$. Using the index theory of a pair of projections [ASS94], we can thus extend the index to operators which are not translation invariant. In higher dimensions (i.e. $d > 1$), we only have a weaker version of this result: The index of U can be expressed using a projection P_L, which depends solely on the interaction length L of U.

Contents

1 Introduction **1**

2 Mathematical Background **3**

3 Quantum Walks and the topological index **6**
3.1 Definition of a Quantum Walk 6
3.2 The Fourier Transform of the operator 8
3.3 The index of a Quantum Walk 10
3.4 A visual interpretation of the index 12

4 Properties of the index **14**
4.1 Class of matrices which define a Quantum Walk 14
4.2 Homotopy invariance 16
4.3 Relation to the bandstructure 16

5 Link to the index of a pair of projections **18**
5.1 One-dimensional case 18
5.2 Higher dimensions 20

6 Summary and Outlook **24**

A Appendix **26**
A.1 The zeros of a multivariate polynomial 26
A.2 Differentiability of the bands 26
A.3 The index of a pair of projections 27
A.4 Proof of Theorem 4 27

Bibliography **31**

1 Introduction

The field of Quantum Walks has experienced increasing popularity in recent years among physicists, mathematicians, and computer scientists. Their study can roughly be split into two different areas of research. On the one hand, Quantum Walks can be seen as the Quantum equivalent of classical Random Walks. Since many classical algorithms can be expressed as Random Walks, it is manifest to assume that the same may hold for Quantum algorithms and Quantum Walks. One such example is the well-known Grover's algorithm, introduced by Grover in 1996 [Gro96]. It is a Quantum search algorithm that manages to reduce the average run time of finding a specific element in an unordered database containing N elements from $O(N)$ to $O(\sqrt{N})$. Further examples of the algorithmic applications of Quantum Walks were given, for instance, by Ambainis [Amb04] and Sarkar et al. [SC19].

On the other hand, Quantum Walks can be seen as modeling the dynamics of a Quantum particle moving on a discrete spatial structure in discrete time. In every time step, the movement of the particle is given by a unitary operator U, which can be traced back to the laws of Quantum Mechanics:

If time were continuous instead of discrete, the dynamics of the particle would be governed by the Schrödinger equation

$$i\,\hbar\,\frac{\partial}{\partial t}\,|\psi(t)\rangle = H\,|\psi(t)\rangle\,, \tag{1.1}$$

where the state of the system is given by $|\psi(t)\rangle$, $\hbar$ is the Planck constant and the self-adjoint operator H is the Hamiltonian of the system. Here, we employ the Dirac notation (also called bra-ket notation), which is properly defined in (2.1). Solving (1.1) for a pure initial state $|\psi(0)\rangle$, the state $|\psi(t)\rangle$ of the system at time t is given by

$$|\psi(t)\rangle = e^{-iHt/\hbar}\;|\psi(0)\rangle\,. \tag{1.2}$$

The unitary operator $U(t) = e^{-iHt/\hbar}$ is defined by the functional calculus and called the time-evolution operator. When considering Quantum Walks, we discretize time and set the operator defining one time step of the walker to be $U = U(1)$. In this sense, a Quantum Walk can be seen as an approximation of the true dynamics of the particle, which are governed by the Schrödinger equation. In particular, Quantum Walks are Quantum Cellular Automata, which can be thought of as describing the movement of multiple (possibly infinitely many) particles on a lattice. Note that some of the index theory developed in this thesis can be generalized to Quantum Cellular Automata, see for instance [Gro+12] and [Vog09].

After revising some of the required mathematical background in chapter 2, we will define a Quantum Walk to be a local, translation invariant, unitary operator on $l^2(\mathbb{Z}^d;\mathbb{C}^s)$. In section 3.2 we will show that the Fourier transform $\hat{U} = \mathcal{F}U\mathcal{F}^{-1}$ is a multiplication operator on $L^2(\mathbb{T}^d;\mathbb{C}^s)$, multiplying a function $\hat{\psi}(k)$ by the matrix $\hat{U}(k)$. The entries of $\hat{U}(k)$ are given by

$$\left(\hat{U}(k)\right)_{n,m} = \sum_{y\in\mathbb{Z}^d} \langle y,n|\,U\,|0,m\rangle\;e^{iy\cdot k}. \tag{1.3}$$

It will turn out that the determinant of $\hat{U}(k)$ is a monomial of the form $C\,e^{iv\cdot k}$ for some $v\in\mathbb{Z}^d$ (see section 3.3). This will enable us to define the index of the Quantum Walk:

$$\mathrm{ind}(U) = v = -i\nabla\left(\log\left(\det\left(\hat{U}(k)\right)\right)\right).$$

We will then go on to derive an alternative formula for $\mathrm{ind}(U)$ in section 3.4:

$$\mathrm{ind}(U) = \sum_{y\in\mathbb{Z}^d} y\ \mathrm{tr}\big(|U_y|^2\big). \tag{1.4}$$

The entries of the matrices $U_y \in \mathbb{C}^{s\times s}$ are given by $(U_y)_{n,m} = \langle y, n|\, U\, |0, m\rangle$. Thus, the matrix U_y represents the information flowing from 0 to $y \in \mathbb{Z}^d$ (see chapter 2 for more details). Using (1.4), we can show that the index corresponds to the average flow of information.

In chapter 4, we will prove some properties of the index. We will start by uniquely characterizing the class of matrices $\hat{U}(k)$ that define Quantum Walks U, which will lead to the additivity of the index, see section 4.1. After this, we will show that the index is a homotopy invariant, using the topology induced by the norm of an operator, see section 4.2. Finally, we will derive yet another formula for the index using the bands $\omega_\alpha(k)$, where $e^{i\omega_\alpha(k)}$ are the eigenvalues of the unitary matrix $\hat{U}(k)$:

$$\mathrm{ind}(U) = \frac{1}{(2\pi)^d} \sum_{\alpha=1}^{s} \int_{\mathbb{T}^d} \nabla\omega_\alpha(k)\, dk.$$

This leads to a connection between the index and the bandstructure of $\hat{U}(k)$. In fact, whenever the gradients of the bands $\omega_\alpha(k)$ are well defined, we have:

$$\mathrm{ind}(U) = \sum_{\alpha=1}^{s} \omega_\alpha(k).$$

The focus of chapter 5 will be on establishing links between the index of a Quantum Walk and the index of a pair of projections, as developed by Avron, Seiler, and Simon [ASS94]. We will show that the index of any local, unitary operator on $l^2(\mathbb{Z};\mathbb{C}^s)$ is given by

$$\mathrm{ind}(U) = \mathrm{tr}\Big(U^\dagger P U - P\Big), \tag{1.5}$$

where P is the projection onto $\mathbb{Z}_{\geq 0}$. This quantity is an integer whenever the trace is well defined, allowing us to generalize our index to one-dimensional Quantum Walks, which are not necessarily translation invariant.

We will show in section 5.2 that for Quantum Walks on the grid $\mathbb{Z}^d$, $d > 1$, no general projection P exists such that a similar formula holds for all U. However, using translation invariance, we can restrict the operator U to a domain $\mathbb{Z}\times\mathbb{T}_L\times\ldots\times\mathbb{T}_L$, where $\mathbb{T}_L$ is a bounded set that depends on the interaction length L of U. The restriction U_L can be viewed as a one-dimensional Quantum Walk, and we have

$$\mathrm{ind}(U)_1 = \frac{1}{(2L+1)^{d-1}}\ \mathrm{tr}\Big(U_L^\dagger P_L U_L - P_L\Big).$$

Hence, there exists a generalization of (1.5) to higher dimensions $d > 1$. The difference, however, is that there exists no general projection P for all operators U since P_L depends on the interaction length of U. Consequently, this does not lead to a definition of the index for every local, unitary operator, unlike in the one-dimensional case. However, as we will argue in the outlook (see Chapter 6), an extension to some classes of operators, specifically ergodic operators, may be possible.

The index of a Quantum Walk was first defined by Kitaev [Kit06] as the flow of a unitary matrix. His works, together with the works of Cedzich [CGW22; Ced+18; Ced+17], Geib [Gei22], Vogts [Gro+12; Vog09] and Asch et al. [ABJ18; ABJ19; AM19] served as inspiration and source material for most of the chapters 3 and 4 and section 5.1. To the best of the author's knowledge, the relation between the index of U and the index of a pair of projections in higher dimensions, as shown in section 5.2, is a new result.

2 Mathematical Background

To be self-contained, we will revise some of the mathematical background in the following section. The setting of the Quantum Walks we will study is a complex Hilbert space $\mathcal{H}$, whose scalar product $\langle\cdot|\cdot\rangle$ we assume to be anti-linear in the 1st and linear in the 2nd coordinate. We employ the Dirac notation (also called bra-ket notation) and denote all elements of $\mathcal{H}$ by $|\psi\rangle$, with the corresponding linear functional being $\langle\psi|$. To be more precise, $\langle\psi|$ is the linear map

$$\langle\psi| : \mathcal{H} \to \mathbb{C},\ \varphi \mapsto \langle\psi\,|\varphi\rangle\,. \tag{2.1}$$

For any linear operator U on $\mathcal{H}$, we denote its adjoint by $U^\dagger$. In this sense, we can write $|\psi\rangle^\dagger = \langle\psi|$. Since we are dealing with Quantum systems, their states will be given by a vector $\psi \in \mathcal{H}$. The state ψ of the system can be interpreted as a complex-valued probability density and is therefore tacitly assumed to be of norm 1.

The Hilbert spaces that we will work with most of the time are $l^2(\mathbb{Z}^d;\mathbb{C}^s)$ and $L^2(\mathbb{T}^d;\mathbb{C}^s)$, where we define $\mathbb{T}^d = \underbrace{\mathbb{S}^1 \times \ldots \times \mathbb{S}^1}_{d-\text{times}}$ to be the d-torus. Their scalar products are given by

$$\langle\psi|\,\varphi\rangle_{l^2(\mathbb{Z}^d;\mathbb{C}^s)} = \sum_{x\in\mathbb{Z}^d} \langle\psi(x)|\,\varphi(x)\rangle_{\mathbb{C}^s} = \sum_{x\in\mathbb{Z}^d}\sum_{j=1}^{s} \overline{\psi(x)_j}\,\varphi(x)_j \quad \text{and}$$

$$\langle f|\,g\rangle_{L^2(\mathbb{T}^d;\mathbb{C}^s)} = \int_{\mathbb{T}^d} \langle f(k)|\,g(k)\rangle_{\mathbb{C}^s}\,dk = \int_{\mathbb{T}^d}\sum_{j=1}^{s} \overline{f_j(k)}\,g_j(k)\,dk$$

Furthermore, for two vector spaces V and W over the same field, we define their tensor product to be the space $V \otimes W$ and associate to it a bilinear map $V \times W \to V \otimes W,\ (v,w) \mapsto v \otimes w$. Elements of the form $v \otimes w$ are called elementary tensors and span the entire space $V \otimes W$. If we are given a basis B_V of V and a basis B_W of W, then the set $\{v \otimes w \mid v \in B_V, w \in B_W\}$ forms a basis of $V \otimes W$. Therefore, the dimension of $V \otimes W$ is the product of the dimensions of V and W.

Due to the universal property, for every bilinear map f from $V \times W$ to some vector space Z, there exists a unique linear map $\tilde{f} : V \otimes W \to Z$, such that $f(v,w) = \tilde{f}(v \otimes w)$ holds for all $v \in V, w \in W$. Thus, there exists an isomorphism between $V \times W$ and $V \otimes W$. In this spirit, we can see the universal property as an alternative definition of the tensor product, i.e. $V \otimes W$ together with the map $\otimes : V \times W \to V \otimes W$ is the unique (up to isomorphisms) vector space such that the universal property holds.

Furthermore, if the spaces V and W are Hilbert spaces with scalar products $\langle\cdot|\cdot\rangle_V$ and $\langle\cdot|\cdot\rangle_W$ respectively, we can define a scalar product $\langle\cdot|\cdot\rangle_{V\otimes W}$ on $V \otimes W$. On the elementary tensors it is given by

$$\langle v_1 \otimes w_1|\,v_2 \otimes w_2\rangle_{V\otimes W} = \langle v_1|\,v_2\rangle_V\ \langle w_1|\,w_2\rangle_W\ \ \forall\, v_1, v_2 \in V, w_1, w_2 \in W$$

and extended by linearity. Additionally, it is easy to see that $l^2(\mathbb{Z}^d) \otimes \mathbb{C}^s \simeq l^2(\mathbb{Z}^d;\mathbb{C}^s)$ and similarly $L^2(\mathbb{T}^d) \otimes \mathbb{C}^s \simeq L^2(\mathbb{T}^d;\mathbb{C}^s)$. Fixing a basis $|1\rangle, \ldots, |s\rangle$ of $\mathbb{C}^s$, an orthonormal basis of $l^2(\mathbb{Z}^d;\mathbb{C}^s)$ is now given by

$$\{\delta_x \otimes |n\rangle\ \mid x \in \mathbb{Z}^d, n = 1, \ldots, s\}.$$

From now on, we will denote elements of this basis by $|x,n\rangle$.

For a family of Hilbert spaces $(\mathcal{H}_i)_{i\in I}$ indexed by some set I we define their direct sum $\mathcal{H} = \bigoplus_{i\in I}\mathcal{H}_i$ to

be all elements $\varphi = (\varphi_i)_{i\in I}$ of the Cartesian product such that $\sum_{i\in I} \|\varphi_i\|^2 < \infty$. The space $\mathcal{H}$ is a Hilbert space with inner product

$$\langle \varphi | \psi \rangle_{\mathcal{H}} = \sum_{i\in I} \langle \varphi_i | \psi_i \rangle_{\mathcal{H}_i}$$

Most of the time, we will assume the Quantum Walks to be translation-invariant (see (2.4)). However, to prove some results for more general operators, we will need to allow a broader class of Hilbert spaces:

We equip every $x \in \mathbb{Z}^d$ with a finite dimensional Hilbert space $\mathcal{H}_x$ and assume that the dimensions of $\mathcal{H}_x$ are uniformly bounded, i.e. $\sup_{x\in\mathbb{Z}^d}(\dim(\mathcal{H}_x)) < \infty$. The space $\mathcal{H}_x$ is called the coin space at position $x \in \mathbb{Z}^d$. We define the separable Hilbert space

$$\mathcal{H} = \bigoplus_{x\in\mathbb{Z}^d} \mathcal{H}_x.$$

For any operator U on $\mathcal{H}$, let $U_{yx} : \mathcal{H}_x \to \mathcal{H}_y$ be the matrix corresponding to the direct sum above. This means that for any $\psi = \bigoplus_{x\in\mathbb{Z}^d} \psi_x \in \mathcal{H}$:

$$U\psi = \bigoplus_{y\in\mathbb{Z}^d} \sum_{x\in\mathbb{Z}^d} U_{yx}\, \psi_x.$$

For every $x \in \mathbb{Z}^d$ we fix an orthonormal basis $|1_x\rangle, ..., |s_x\rangle$ of $\mathcal{H}_x$. The entries of U_{yx} with respect to these bases are given by $\left(U_{yx}\right)_{n,m} = \left\langle y, n_y \middle| U \middle| x, m_x \right\rangle$. An operator U on $\mathcal{H}$ is called local if there exists some $L \in \mathbb{N}$ such that

$$U_{yx} = 0 \text{ if } |y - x| > L. \tag{2.2}$$

Here, $|\cdot|$ denotes some arbitrary norm on $\mathbb{Z}^d$. The constant L is called the interaction length of U.

If $\mathcal{H}_x = \mathbb{C}^s$ for every $x \in \mathbb{Z}^d$, then $\mathcal{H} \simeq l^2(\mathbb{Z}^d; \mathbb{C}^s)$. We fix an orthonormal basis $|1\rangle, ..., |s\rangle$ of $\mathbb{C}^s$ and define for every $v \in \mathbb{Z}^d$ the translation operator

$$T_v : \mathcal{H} \to \mathcal{H},\ |x, n\rangle \mapsto |x + v, n\rangle\,. \tag{2.3}$$

It is easily verified that its adjoint $T_v^\dagger$ is just T_{-v}. To check this, we compute for all elements of the orthonormal basis $|x, n\rangle, |y, m\rangle$:

$$\langle y, m | T_v | x, n\rangle = \langle y, m | x + v, n\rangle = 1 \iff y = x + v \text{ and } n = m$$
$$\langle \Big(T_{-v} | y, m\rangle \Big) | x, n\rangle = \langle y - v, m | x, n\rangle = 1 \iff y - v = x \text{ and } n = m$$

We call an operator U on $\mathcal{H}$ translation invariant, if it commutes with all translation operators T_v, i.e.

$$[U, T_v] = UT_v - T_vU = 0 \ \ \forall v \in \mathbb{Z}^d. \tag{2.4}$$

Note that any operator T_v can be expressed as a finite composition of the operators $T_{e_1}, ..., T_{e_d}$, where $e_1, ..., e_d$ is the standard basis of $\mathbb{R}^d$. Hence, it suffices to check whether the operator U commutes with the operators $T_{e_1}, ..., T_{e_d}$. If U is assumed to be translation invariant, this leads to:

$$\left(U_{yx}\right)_{n,m} = \langle y, n | U T_x | 0, m\rangle = \langle y - x, n | U | 0, m\rangle\,.$$

Thus, the matrix U_{yx} depends solely on the difference $y - x$. To simplify the notation, we therefore define the matrix $U_y = U_{y0} : \mathbb{C}^s \to \mathbb{C}^s$. Its entries are given by

$$\left(U_y\right)_{n,m} = \langle y, n | U | 0, m\rangle\,. \tag{2.5}$$

Remark 1. *We can also consider a more general form of translation invariance. Instead of commuting with every shift operator T_v we can require U to commute only with the specific multiples $T_{n_1 e_1}, ..., T_{n_d e_d}$ of the standard shifts. Thus, if U commutes with $T_{n_i e_i}$, then it is n_i-periodic in direction e_i. However, these two definitions are equivalent: By considering each group of n_i cells to be a single cell we change the coin space from $\mathbb{C}^s$ to $\mathbb{C}^{n_i s}$. The operator U considered as an operator on $l^2(\mathbb{Z}^d; \mathbb{C}^{n_i s})$ now commutes with T_{e_i}. Repeating this restructuring for all directions e_i shows the equivalence of the two definitions.*

Definition 1 (Fourier Transform)**.** *For an element $\psi \in l^2(\mathbb{Z}^d; \mathbb{C})$ we define its Fourier Transform $\hat{\psi} \in L^2(\mathbb{T}^d; \mathbb{C})$ to be*

$$\hat{\psi}(k) = \frac{1}{\sqrt{2\pi}^d} \sum_{x \in \mathbb{Z}^d} \psi(x)\, e^{ik \cdot x}.$$

The Fourier Transform $\mathcal{F} : l^2(\mathbb{Z}^d; \mathbb{C}) \to L^2(\mathbb{T}^d; \mathbb{C})$, $\psi \mapsto \hat{\psi}$ is bijective with inversion

$$\psi(x) = \frac{1}{\sqrt{2\pi}^d} \int_{\mathbb{T}^d} \hat{\psi}(k)\, e^{-ik \cdot x}\, dk.$$

Defined this way, the Fourier Transform is a unitary operator between Hilbert spaces. On $l^2(\mathbb{Z}^d; \mathbb{C}^s)$ it acts on each component separately, i.e.

$$\mathcal{F} : l^2(\mathbb{Z}^d; \mathbb{C}^s) \to L^2(\mathbb{T}^d; \mathbb{C}^s), \quad \begin{pmatrix} \psi_1 \\ \vdots \\ \psi_s \end{pmatrix} \mapsto \begin{pmatrix} \hat{\psi}_1 \\ \vdots \\ \hat{\psi}_s \end{pmatrix}.$$

3 Quantum Walks and the topological index

In this chapter, we will first specify the notion of Quantum Walk using its operator U. In the next step, we will show that its Fourier Transform $\hat{U}$ is a multiplication operator that multiplies the state $\hat{\psi}(k)$ by the matrix $\hat{U}(k) \in \mathbb{C}^{s\times s}$. We will then show that the determinant of the matrix $\hat{U}(k)$ is a Laurent polynomial and define its exponent to be the index of U. At the end of every section, we will illustrate the results on the hexagonal lattice.

3.1 Definition of a Quantum Walk

We consider the separable Hilbert space $\mathcal{H} = l^2(\mathbb{Z}^d; \mathbb{C}^s)$, where d is the dimension of the lattice and s the dimension of the coin space. A Quantum Walk is defined as follows:

Definition 2 (Quantum Walk)**.** *A Quantum Walk is a local, translation invariant, unitary operator on $\mathcal{H}$.*

For a thorough definition of locality and translation invariance see (2.2) and (2.4). Intuitively, requiring U to be translation invariant means that the dynamics of the walker are independent of its location on the grid. Using the same intuition, locality entails that in one time step the walker can only reach sites that are sufficiently close to its current location. These two conditions are not necessarily required to define Quantum Walks in a meaningful way. However, they are fundamental for the majority of the results in this thesis. Thus, whenever we talk of Quantum Walks from now on, we will assume them to be local, translation invariant, and unitary unless otherwise specified. Nevertheless, much of the work that served as inspiration and source for this thesis deals with one-dimensional Quantum Walks that are not translation invariant, such as [Gro+12; Vog09], or not strictly local [ABJ18; Gei22].

Proposition 1. *The set of Quantum Walks (i.e. local, translation invariant, unitary operators on $l^2(\mathbb{Z}^d; \mathbb{C}^s)$) forms a group, its group operation being the composition of operators.*

Proof. We have for two translation invariant operators U and V and every shift T_v:

$$[UV, T_v] = UVT_v - T_vUV = U(VT_v - T_vV) = 0.$$

Thus, the composition of two translation invariant operators stays translation invariant. The same obviously holds for unitarity and locality. Hence, the composition of Quantum Walks is still a Quantum Walk.

Furthermore, the inverse of any Quantum Walk U is its adjoint $U^\dagger$, which is again local and unitary. Additionally, using $T_v^\dagger = T_{-v}$, we compute for every shift T_v

$$[U^\dagger, T_v] = U^\dagger T_v - T_vU^\dagger = U^\dagger T_{-v}^\dagger - T_{-v}^\dagger U^\dagger = (T_{-v}U)^\dagger - (UT_{-v})^\dagger = [T_{-v}, U]^\dagger = 0.$$

Thus, the set of Quantum Walk is stable under the inversion of operators. Since the identity can trivially be viewed as a Quantum Walk, the set of all Quantum Walks forms a group. □

Example 1 (Quantum Walks on the hexagonal lattice)**.** We define the hexagonal lattice $\Gamma = \Gamma_A \cup \Gamma_B$ (see Figure 3.1) as the union of the two lattices Γ_A and Γ_B, where

$$\Gamma_A = \operatorname{span}_{\mathbb{Z}}\left(\begin{pmatrix}\sqrt{3}\\0\end{pmatrix}, \frac{1}{2}\begin{pmatrix}\sqrt{3}\\3\end{pmatrix}\right) \quad \text{and} \quad \Gamma_B = \Gamma_A + \begin{pmatrix}0\\1\end{pmatrix}.$$

Furthermore, we define the Hilbert space

$$l^2(\Gamma) = \left\{ \sum_{z\in\Gamma} \psi_z |z\rangle \text{ st. } \psi_z \in \mathbb{C}, \sum_{z\in\Gamma} |\psi_z|^2 < \infty \right\} \simeq l^2(\mathbb{Z}^2;\mathbb{C}^2).$$

Using the three-dimensional coin space $\mathbb{C}^3$, we set the Hilbert space of the Quantum Walk to be

$$\mathcal{H} = l^2(\Gamma) \otimes \mathbb{C}^3 \simeq l^2(\mathbb{Z}^2;\mathbb{C}^6).$$

An intuitive way to picture a Quantum Walk (and one that is also used in many physical models) is considering it to be a composition of shift and coin operators. As the name suggests, the shift operator S performs a shift in the lattice, only acting on $l^2(\Gamma)$ and leaving the coin space $\mathbb{C}^3$ unchanged. On the other hand, the coin operator C acts only on the coin space.

The shift operator performs a different shift depending on the state of the coin space. Since the coin space is three-dimensional, we define three different shifts S_1, S_2 and S_3, which are operators on $l^2(\mathbb{Z}^2;\mathbb{C}^2)$ and perform a shift $|x, i\rangle \mapsto |x + v, j\rangle$. Note that $i = j$ corresponds to staying in the same sublattice Γ_A or Γ_B. For instance, one way to define the shifts in terms of the basis $|(x_1, x_2), i\rangle$ is:

$$\begin{aligned}
S_1(|(x_1, x_2), 1\rangle) &= |(x_1, x_2 - 1), 2\rangle \\
S_1(|(x_1, x_2), 2\rangle) &= |(x_1 - 1, x_2 + 1), 1\rangle \\
S_2(|(x_1, x_2), 1\rangle) &= |(x_1 + 1, x_2 - 1), 2\rangle \\
S_2(|(x_1, x_2), 2\rangle) &= |(x_1, x_2), 1\rangle \\
S_3(|(x_1, x_2), 1\rangle) &= |(x_1, x_2), 2\rangle \\
S_3(|(x_1, x_2), 2\rangle) &= |(x_1, x_2 + 1), 1\rangle .
\end{aligned}$$

The action of the shifts S_1, S_2 and S_3 is illustrated in Figure 3.1 below:

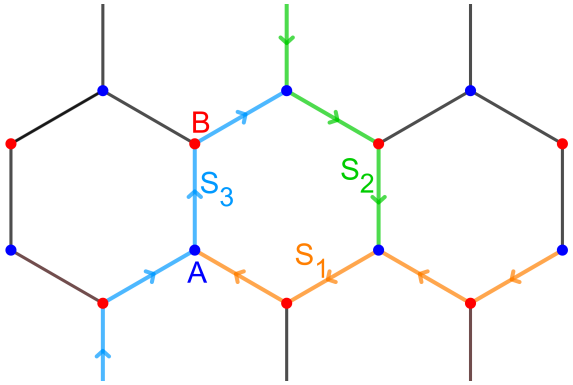


Figure 3.1 The shifts on the hexagonal lattice

Fixing an orthonormal basis $|a\rangle, |b\rangle, |c\rangle$ of $\mathbb{C}^3$, the shift operator S is now defined to be

$$S = S_1 \otimes |a\rangle\langle a| + S_2 \otimes |b\rangle\langle b| + S_3 \otimes |c\rangle\langle c| .$$

Let $C_A, C_B \in U(3)$ be unitary matrices. We set the coin operator C on $\mathcal{H}$:

$$C(|x, i\rangle \otimes \tau) = \begin{cases} |x, i\rangle \otimes C_A\tau & \text{if } i = 1 \\ |x, i\rangle \otimes C_B\tau & \text{if } i = 2 \end{cases},$$

where $\tau \in \{|a\rangle, |b\rangle, |c\rangle\}$. The Quantum Walk is now given by the unitary operator $U = SC$.

Remark 2. *The above construction is an example of a shift-coin decomposition of a Quantum Walk. It is an interesting question whether any Quantum Walk can be decomposed into a finite sequence of shifts and coins. The answer to this certainly depends on our definition of a Quantum Walk and the dimension of the underlying lattice. For example, for one-dimensional walks (i.e. walks on $\mathbb{Z}$), it has been shown that any local, unitary operator can be decomposed into a sequence of elementary shifts and coins [CGW22].*

Efforts have been made to obtain similar results in higher dimensions, for example, in the two-dimensional case along the lines of [MLK77] and [KV90]. However, so far no similar statements have been found, and it appears that the factorization holds only in the one-dimensional case [Vog09].

3.2 The Fourier Transform of the operator

For a Quantum Walk U, we define its Fourier Transform $\hat{U} = \mathcal{F}U\mathcal{F}^{-1}$, where $\mathcal{F}$ is defined as in Definition 1. $\hat{U}$ is an operator on $L^2(\mathbb{T}^d;\mathbb{C}^s)$ and, using translation invariance, we can show the following (see for instance [Kit06]):

Proposition 2. *$\hat{U}$ is a multiplication operator by the matrix $\hat{U}(k)$. In other words, we have for $\hat{\psi} \in L^2(\mathbb{T}^d;\mathbb{C}^s)$ and $k \in \mathbb{T}^d$:*

$$\left(\hat{U}\,\hat{\psi}\right)(k) = \hat{U}(k)\,\hat{\psi}(k).$$

The entries of the matrix $\hat{U}(k) \in \mathbb{C}^{s\times s}$ are given by

$$\left(\hat{U}(k)\right)_{n,m} = \sum_{y\in\mathbb{Z}^d} \langle y,n|\,U\,|0,m\rangle\; e^{iy\cdot k}, \tag{3.1}$$

where only finitely many terms are non-zero.

Note that the coefficients in (3.1) correspond to the entries of U_y as defined in (2.5). Thus, we have

$$\hat{U}(k) = \sum_{y\in\mathbb{Z}^d} U_y\, e^{iy\cdot k}. \tag{3.2}$$

Proof. Using $\langle x,n|\,\psi\rangle = \psi_n(x)$ for $\psi \in l^2(\mathbb{Z}^d;\mathbb{C}^s)$, we compute for fixed $n \in \{1,...,s\}$:

$$\begin{aligned}(U\psi(x))_n &= \langle x,n|\,U\,|\psi\rangle = \langle x,n|\,UI\,|\psi\rangle = \langle x,n|\,U\left(\sum_{m=1}^{s}\sum_{y\in\mathbb{Z}^d}|y,m\rangle\langle y,m|\right)|\psi\rangle \\ &= \sum_{m=1}^{s}\sum_{y\in\mathbb{Z}^d}\langle x,n|\,U\,|y,m\rangle\langle y,m|\,\psi\rangle,\end{aligned} \tag{3.3}$$

where we used continuity (and linearity) of the scalar product. Using translation invariance of U and $T_v^\dagger = T_{-v}$, we verify

$$\langle x,n|\,U\,|y,m\rangle = \langle 0,n|\,T_x^\dagger U\,|y,m\rangle = \langle 0,n|\,UT_{-x}\,|y,m\rangle = \langle 0,n|\,U\,|y-x,m\rangle\,. \tag{3.4}$$

Similarly, it holds that

$$\langle x,n|\,U\,|y,m\rangle = \langle x-y,n|\,U\,|0,m\rangle\,. \tag{3.5}$$

With this, we continue in (3.3):

$$\begin{aligned}(U\psi(x))_n &= \sum_{m=1}^{s}\sum_{y\in\mathbb{Z}^d}\langle x,n|\,U\,|y,m\rangle\langle y,m|\,\psi\rangle = \sum_{m=1}^{s}\sum_{y\in\mathbb{Z}^d}\langle 0,n|\,U\,|y-x,m\rangle\langle y,m|\,\psi\rangle \\ &= \sum_{m=1}^{s}\sum_{y\in\mathbb{Z}^d}\langle 0,n|\,U\,|y,m\rangle\langle y+x,m|\,\psi\rangle = \sum_{m=1}^{s}\sum_{y\in\mathbb{Z}^d}\langle 0,n|\,U\,|y,m\rangle\;\psi_m(y+x) \\ &= \sum_{m=1}^{s}\sum_{y\in\mathbb{Z}^d}\langle 0,n|\,U\,|y,m\rangle\;\left(T_y\psi(x)\right)_m\,.\end{aligned}$$

Using $\mathcal{F}T_v\psi(k) = e^{-iv\cdot k}\hat{\psi}(k)$ we compute the Fourier Transform at $k \in \mathbb{T}^d$

$$\mathcal{F}(U\psi)_n(k) = \sum_{m=1}^{s}\sum_{y\in\mathbb{Z}^d} \langle 0,n|\, U\,|y,m\rangle\; e^{-iy\cdot k}\hat{\psi}_m(k) = \sum_{m=1}^{s}\sum_{y\in\mathbb{Z}^d} \langle 0,n|\, U\,|-y,m\rangle\; e^{iy\cdot k}\hat{\psi}_m(k)$$
$$\overset{(3.5)}{=} \sum_{m=1}^{s}\sum_{y\in\mathbb{Z}^d} \langle y,n|\, U\,|0,m\rangle\; e^{iy\cdot k}\hat{\psi}_m(k) = \left(\hat{U}(k)\hat{\psi}(k)\right)_n,$$

where the entries of $\hat{U}(k) \in \mathbb{C}^{s\times s}$ are given by:

$$\left(\hat{U}(k)\right)_{n,m} = \sum_{y\in\mathbb{Z}^d} \langle y,n|\, U\,|0,m\rangle\; e^{iy\cdot k}.$$

Since U is a local operator, only finitely many $y \in \mathbb{Z}^d$ lead to non-zero terms. □

We also have the following straightforward Corollary:

Corollary 1. *The matrix $\hat{U}(k)$ is unitary and its inverse $\hat{U}(k)^\dagger$ is the matrix corresponding to the Quantum Walk $U^\dagger$.*

Proof. The set of Quantum Walks forms a group, as seen in Proposition 1. Hence, the adjoint $U^\dagger$ also describes a Quantum Walk. Thus, the Fourier Transform of $U^\dagger$ is a multiplication by some matrix $\tilde{U}(k)$, whose entries are given by:

$$\left(\tilde{U}(k)^\dagger\right)_{n,m} = \overline{\left(\tilde{U}(k)\right)_{m,n}} = \overline{\sum_{y\in\mathbb{Z}^d} \langle y,m|\, U^\dagger\,|0,n\rangle\; e^{iy\cdot k}} = \sum_{y\in\mathbb{Z}^d} \langle 0,n|\, U\,|y,m\rangle\; e^{-iy\cdot k}$$
$$= \sum_{y\in\mathbb{Z}^d} \langle 0,n|\, U\,|-y,m\rangle\, e^{iy\cdot k} \overset{(3.5)}{=} \sum_{y\in\mathbb{Z}^d} \langle y,n|\, U\,|0,m\rangle\, e^{iy\cdot k} = \left(\hat{U}(k)\right)_{n,m}.$$

We conclude that $\tilde{U}(k) = \hat{U}(k)^\dagger$ holds. Furthermore, we compute for every $\psi \in l^2(\mathbb{Z}^d;\mathbb{C}^s)$:

$$\hat{U}(k)\tilde{U}(k)\hat{\psi}(k) = \mathcal{F}U\mathcal{F}^{-1}\mathcal{F}U^\dagger\mathcal{F}^{-1}\hat{\psi}(k) = \mathcal{F}UU^\dagger\mathcal{F}^{-1}\hat{\psi}(k) = \hat{\psi}(k).$$

Hence, $\hat{U}(k)\hat{U}(k)^\dagger = I$ and $\hat{U}(k)$ is unitary. □

Example 2. (Quantum Walks on the hexagonal lattice continued) Continuing Example 1, we explicitly compute the matrix $\hat{U}(k)$.
We define the Fourier Transform of the shifts $\hat{S}_1 = \mathcal{F}S_1\mathcal{F}^{-1}$ and $\hat{S}_2, \hat{S}_3$ similarly. We compute for every $\psi \in l^2(\mathbb{Z}^2;\mathbb{C}^2)$:

$$S_1\psi = S_1\Big(\sum_{x\in\mathbb{Z}^2} |x,1\rangle\; \psi_1(x) + |x,2\rangle\; \psi_2(x)\Big) = \sum_{x\in\mathbb{Z}^2} |(x_1,x_2-1),2\rangle\; \psi_1(x) + |(x_1-1,x_2+1),1\rangle\; \psi_2(x).$$

Therefore, the output $S_1\psi$ is given by

$$(S_1\psi)_1(x_1,x_2) = \psi_2(x_1+1,x_2-1) = T_{(1,-1)}\psi_2(x) \quad \text{and}$$
$$(S_1\psi)_2(x_1,x_2) = \psi_1(x_1,x_2+1) = T_{(0,1)}\psi_1(x).$$

Since the Fourier Transform acts on each coordinate separately and using $\mathcal{F}T_v\psi(k) = e^{-iv\cdot k}\hat{\psi}(k)$, this leads to:

$$\hat{S}_1\hat{\psi}(k) = \big(\mathcal{F}S_1\psi\big)(k) = \left(\mathcal{F}\begin{pmatrix} T_{(1,-1)}\psi_2 \\ T_{(0,1)}\psi_1\end{pmatrix}\right)(k) = \begin{pmatrix} e^{i(k_2-k_1)}\hat{\psi}_2(k) \\ e^{-ik_2}\hat{\psi}_1(k)\end{pmatrix} = \begin{pmatrix} 0 & e^{i(k_2-k_1)} \\ e^{-ik_2} & 0\end{pmatrix}\begin{pmatrix}\hat{\psi}_1(k)\\ \hat{\psi}_2(k)\end{pmatrix}.$$

Analogously we compute

$$\hat{S}_2\,\hat{\psi}(k) = \begin{pmatrix} 0 & 1 \\ e^{i(k_1-k_2)} & 0 \end{pmatrix} \hat{\psi}(k) \quad \text{and} \quad \hat{S}_3\hat{\psi}(k) = \begin{pmatrix} 0 & e^{ik_2} \\ 1 & 0 \end{pmatrix} \hat{\psi}(k).$$

We notice that the coefficients of k_1 and k_2 are exactly the same as the shifts used to define S_1, S_2 and S_3. The Fourier transform of the shift operator is now given by

$$\hat{S}\,\hat{\psi}(k) = \begin{pmatrix} 0 & 0 & 0 & e^{i(k_2-k_1)} & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & e^{ik_2} \\ e^{-ik_2} & 0 & 0 & 0 & 0 & 0 \\ 0 & e^{i(k_1-k_2)} & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{pmatrix} \hat{\psi}(k).$$

Finally, it follows that $\hat{U}$ is given as a multiplication operator of the form

$$\hat{U}\hat{\psi}(k) = \begin{pmatrix} 0 & S_{BA}(k)\,C_B \\ S_{AB}(k)\,C_A & 0 \end{pmatrix} \hat{\psi}(k) =: \hat{U}(k)\,\hat{\psi}(k),$$

where $S_{AB}(k) = \begin{pmatrix} e^{-ik_2} & 0 & 0 \\ 0 & e^{i(k_1-k_2)} & 0 \\ 0 & 0 & 1 \end{pmatrix}$ and $S_{BA}(k) = \begin{pmatrix} e^{i(k_2-k_1)} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & e^{ik_2} \end{pmatrix}$.

This is in accordance with equation (3.1), which would have allowed us to compute the entries of $\hat{U}(k)$ directly. We also remark that if the shift were to act only within each sublattice Γ_A and Γ_B, the resulting block matrix $\hat{U}(k)$ would be diagonal instead of off-diagonal.

3.3 The index of a Quantum Walk

The aim of this section is proving the following statement about the determinant of $\hat{U}(k)$, which was for instance already shown by [Gro+12; Vog09]:

Theorem 1. *The determinant of $\hat{U}(k)$ is a monomial, i.e.* $\det\left(\hat{U}(k)\right) = C\,e^{iv\cdot k}$ *for some $v \in \mathbb{Z}^d$ and $C \in \mathbb{C}$ with $|C| = 1$.*

The proof of the theorem relies on the following algebraic Lemma:

Lemma 1. *Let f, g be two Laurent polynomials such that $f(z)g(z) = 1$ holds on $\{z \in \mathbb{C}^d \,|\, \forall\, i :\ z_i \neq 0\}$. Then both f and g are monomials.*

The idea for the proof of this Lemma originates from a discussion on the forum StackExchange and has only been slightly modified for our purposes [Som23].

Proof. Let $M \in \mathbb{N}$ be sufficiently big such that $\tilde{f}(z) = z^M f(z)$ is a polynomial. Note that here z^M stands for $z_1^M \cdot \ldots \cdot z_d^M$, whereas z^n for $n \in \mathbb{Z}^d$ stands for $z_1^{n_1} \cdot \ldots \cdot z_d^{n_d}$. We want to prove that $\tilde{f}$ and therefore f are monomials.

Let $\mathcal{A} = \{z \in \mathbb{C}^d \ :\ z_i = 0 \text{ for some } i\}$ be the axes of $\mathbb{C}^d$. Since f and g are Laurent polynomials, their poles can only be in the set $\mathcal{A}$. Using $f(z)g(z) = 1$ for all $z \notin \mathcal{A}$, we get that also the zeros of f and g must be in $\mathcal{A}$. Thus, if $\tilde{f}(z) = 0$, then $z \in \mathcal{A}$.

Suppose that $\tilde{f}$ is not a monomial, i.e. it is not of the form $\tilde{f}(z) = Cz^n$ for some $C \in \mathbb{C}$ and $n \in \mathbb{N}^d$. Then there exists an index $i \in \{1, \ldots, d\}$ and $n, n' \in \mathbb{N}^d$ such that $n_i \neq n_i'$ and the monomials z^n and $z^{n'}$ have non-zero coefficients in $\tilde{f}$. To simplify the notation, we may assume without loss of generality that $i = 1$,

$n_1 < n_1'$ and that all other exponents of z_1 in $\tilde{f}$ are between n_1 and n_1'. Denoting $k = n_1$ and $m = n_1' - n_1$, we can thus write

$$\tilde{f}(z) = z_1^k \left(z_1^m \, p_m + z_1^{m-1} \, p_{m-1} + ... + p_0 \right),$$

where the p_j are polynomials in the variables $z_2, ..., z_d$ and $p_m \neq 0, p_0 \neq 0$ (i.e. they are not the zero polynomials, but they may have points where they evaluate to 0). Fixing these $d-1$ variables, we define the polynomial

$$f_1 : \mathbb{C} \to \mathbb{C},\ x \mapsto x^k \underbrace{\left(x^m \, p_m + x^{m-1} \, p_{m-1} + ... + p_0 \right)}_{=p(x)}.$$

Here, p is a univariate polynomial in $x \in \mathbb{C}$. Suppose we had $(z_2, ..., z_d) \in \mathbb{C}^{d-1}$ such that $z_j \neq 0$ for all $j \geq 2$, $p_m(z_2, ..., z_d) \neq 0$ and $p_0(z_2, ..., z_d) \neq 0$. Then, since $p_m \neq 0$ and $p_0 \neq 0$, a nonzero root of p would exist. Denoting this root by z_1 and defining the point $z^* = (z_1, ..., z_d)$, we would have $\tilde{f}(z^*) = f_1(z_1) = 0$ and $z_j^* \neq 0$ for all $j \in \{1, ..., d\}$. In other words, z^* is a root of $\tilde{f}$, which is not in $\mathcal{A}$, a contradiction.

However, the existence of such a point $(z_2, ..., z_d) \in \mathbb{C}^{d-1}$ is obvious: Since p_m and p_0 are non-zero polynomials, their zero-sets are sets of measure 0. Additionally, the set $\mathcal{A}_{d-1} = \{z \in \mathbb{C}^{d-1} \text{ st } z_j = 0 \text{ for some } j\}$ has measure zero as well. Hence the union of those three sets has measure zero in $\mathbb{C}^{d-1}$, which is why its complement is non-empty, proving the existence of $(z_2, ..., z_d)$.
Thus, the assumption that $\tilde{f}$ is not a monomial leads to a contradiction, finishing the proof. □

Now we are ready to prove Theorem 1:

Proof. We use the Leibniz rule to compute

$$\begin{aligned} \det\left(\hat{U}(k)\right) &= \sum_{\sigma \in S_s} \operatorname{sign}(\sigma) \prod_{n=1}^{s} \left(\hat{U}(k)\right)_{n,\sigma(n)} = \sum_{\sigma \in S_s} \operatorname{sign}(\sigma) \prod_{n=1}^{s} \left(\sum_{y \in \mathbb{Z}^d} \langle y, n | \, U \, | 0, \sigma(n) \rangle \; e^{iy \cdot k} \right) \\ &= \sum_{j=1}^{J} C_j \, e^{i y^{(j)} \cdot k}, \end{aligned}$$

where $C_j \in \mathbb{C} \backslash \{0\}$ and $y^{(j)} \in \mathbb{Z}^d$. The last equality holds since only finitely many $y \in \mathbb{Z}^d$ lead to non-zero terms. Thus, $\det(\hat{U}(k))$ is a Laurent polynomial in the variables $e^{ik_1}, ...e^{ik_d}$.
Let $f : \mathbb{C}^d \to \mathbb{C},\ z \mapsto \sum_{j=1}^{J} C_j \, z^{y^{(j)}}$ be its extension onto $\mathbb{C}^d$, where $z^y = z_1^{y_1} \cdot ... \cdot z_d^{y_d}$. By relabeling we can write

$$f(z) = \sum_{n \in [-M,N]^d \cap \mathbb{Z}^d} C_n z^n,$$

where $M, N \in \mathbb{Z}$ such that $-M \leq N$. We can repeat the same construction for $U^\dagger$ to get another Laurent polynomial

$$g : \mathbb{C}^d \to \mathbb{C},\ z \mapsto \sum_{n \in [-M',N']^d \cap \mathbb{Z}^d} D_n \, z^n.$$

Since f and g represent the determinants of U and $U^\dagger$, we have $f(z)\, g(z) = 1$ for all $z \in \mathbb{S}^d$. We define the polynomial h in d variables over the field $\mathbb{C}$:

$$h : \mathbb{C}^d \to \mathbb{C},\ z \mapsto \left(f(z) g(z) - 1 \right) z^{M+M'}.$$

Since $h(z_1, ..., z_d) = 0$ for all $z_1, ..., z_d \in \mathbb{S}^1$, we use Corollary 4 from the Appendix, a proof of which can be found in [Lan02], to conclude $h = 0$. This means that $f(z)\, g(z) = 1$ for all $z \in \mathbb{C}^d$ such that $z_i \neq 0$ for all $i \in \{1, ..., d\}$. Using Lemma 1 finishes the proof. □

This allows us to define the index of a Quantum Walk:

Definition 3. *The index ind(U) of a Quantum Walk U is the unique $v \in \mathbb{Z}^d$ such that* $\det\left(\hat{U}(k)\right) = C\, e^{iv\cdot k}$.

Note that this is not the only topological index one can define for a Quantum Walk, see for example [Gei22; Sta18] for other indices. Nevertheless, we will from now on call it *the* index of a Quantum Walk.

Example 3. (Quantum Walks on the hexagonal lattice continued) We saw in Example 2 that if the operator U is assumed to have a shift-coin decomposition $U = SC$, with shift S and coin C defined as in Example 1, then the matrix $\hat{U}(k)$ is of the form

$$\hat{U}(k) = \begin{pmatrix} 0 & S_{BA}(k)C_B \\ S_{AB}(k)C_A & 0 \end{pmatrix},$$

where $C_A, C_B \in U(3)$ are the unitary coin matrices and $S_{AB}(k), S_{BA}(k)$ are diagonal matrices with entries $e^{iy\cdot k}$ on the diagonal. Due to the blockstructure of $\hat{U}(k)$, we can compute its determinants

$$\det\left(\hat{U}(k)\right) = \det(-S_{BA}(k)\, C_B\, S_{AB}(k)\, C_A) = (-1)^d\, \det(S_{BA}(k))\, \det(C_B)\, \det(S_{AB}(k))\, \det(C_A),$$

where we used the multiplicativity of the determinant. Since the shift matrices are diagonal, we can easily compute their determinant

$$\det(S_{AB}(k)) = e^{-ik_1} \quad \text{and} \quad \det(S_{BA}(k)) = e^{ik_1}.$$

Hence their determinants cancel out, and the determinant of $\hat{U}(k)$ is the product of the determinants of C_A and C_B. Therefore $\det\left(\hat{U}(k)\right)$ is independent of k and the index of U is 0.

As we saw above, the index depends only on the determinants of the shift matrices S_{AB} and S_{BA}. However, we know from equation (3.1) that the entries of the shift matrices depend only on the shifts used to define them. In other words, a shift from 0 to $y \in \mathbb{Z}^2$ corresponds to an entry of $e^{ik\cdot y}$ on the diagonal of one of the shift matrices. Thus, the index of U is just the sum of the six shifts we use in its definition. This is a specific instance of a more general fact: The index corresponds to the average flow of information, as we will show in the following section.

3.4 A visual interpretation of the index

In this section, we will derive an alternative way to compute the index and use it to give a visual interpretation of its effect. This will additionally explain why it is called the information flow index by some authors [see for example Gei22; Vog09]. The following computations are along the lines of [Gro+12].

As shown in Section 3.3, the determinant of $\hat{U}(k)$ is of the form $C\, e^{ik\cdot v}$ and we define its index to be $v \in \mathbb{Z}^d$. We compute:

$$\operatorname{ind}(U)_j = v_j = \frac{1}{(2\pi)^d i} \int_{\mathbb{T}^d} iv_j\, dk = \frac{1}{(2\pi)^d i} \int_{\mathbb{T}^d} \frac{1}{\det\left(\hat{U}(k)\right)} \frac{\partial}{\partial k_j} \det\left(\hat{U}(k)\right) dk. \tag{3.6}$$

In order to compute the derivative of $\det\left(\hat{U}(k)\right)$ we use Jacobi's formula: Let $t \mapsto A(t)$ be a differentiable map from the real numbers to the invertible square matrices, then the derivative of $\det(A(t))$ is given by

$$\frac{d}{dt} \det(A(t)) = \det(A(t))\, \operatorname{tr}\left(A(t)^{-1} \frac{d}{dt} A(t)\right),$$

where in $\frac{d}{dt}A(t)$ the derivative is taken in each entry of $A(t)$. A proof of Jacobi's formula can, for instance, be found in [MN99, Theorem 8.1]. Using this we continue from equation (3.6):

$$\operatorname{ind}(U)_j = \frac{1}{(2\pi)^d i} \int_{\mathbb{T}^d} \operatorname{tr}\left(\hat{U}(k)^\dagger \frac{\partial}{\partial k_j} \hat{U}(k)\right) dk. \tag{3.7}$$

We recall equation (3.2) and compute:

$$\frac{\partial}{\partial k_j}\hat{U}(k) = \sum_{y\in\mathbb{Z}^d} iy_j\, U_y e^{ik\cdot y}.$$

This means that

$$\hat{U}(k)^\dagger \frac{\partial}{\partial k_j}\hat{U}(k) = \left(\sum_{x\in\mathbb{Z}^d} U_x^\dagger\, e^{-ik\cdot x}\right)\left(\sum_{y\in\mathbb{Z}^d} iy_j\, U_y e^{ik\cdot y}\right) = \sum_{y,x\in\mathbb{Z}^d} i\, y_j\, e^{ik\cdot(y-x)}\, U_x^\dagger U_y.$$

Plugging this into (3.7) gives:

$$\begin{aligned}\mathrm{ind}(U)_j &= \frac{1}{(2\pi)^d i}\int_{\mathbb{T}^d}\sum_{x,y\in\mathbb{Z}^d} i\, y_j e^{ik\cdot(y-x)}\ \mathrm{tr}\Big(U_x^\dagger U_y\Big)\, dk = \sum_{x,y\in\mathbb{Z}^d} y_j\ \mathrm{tr}\Big(U_x^\dagger U_y\Big)\,\delta_{x,y}\\ &= \sum_{y\in\mathbb{Z}^d} y_j\,\mathrm{tr}\big(|U_y|^2\big).\end{aligned}$$

Thus, the index is given by

$$\mathrm{ind}(U) = \sum_{y\in\mathbb{Z}^d} y\ \mathrm{tr}\big(|U_y|^2\big). \tag{3.8}$$

We want to interpret this formula in order to give a visual meaning to the index. Thus, we compute:

$$\begin{aligned}\mathrm{tr}\big(|U_y|^2\big) = \mathrm{tr}\Big(U_y^\dagger U_y\Big) &= \sum_{m=1}^{s}(U_y^\dagger U_y)_{m,m} = \sum_{n,m=1}^{s}(U_y^\dagger)_{m,n}\,(U_y)_{n,m} = \sum_{n,m=1}^{s}\overline{(U_y)_{n,m}}\,(U_y)_{n,m} = \sum_{n,m=1}^{s}|(U_y)_{n,m}|^2\\ &= \sum_{n,m=1}^{s}|\,\langle y,n|\,U\,|0,m\rangle\,|^2.\end{aligned}$$

In this sense $\mathrm{tr}\big(|U_y|^2\big)$ measures the total amount of information moving from $0\in\mathbb{Z}^d$ to $y\in\mathbb{Z}^d$ averaged over the coin space. Additionally, Parseval's identity gives

$$\begin{aligned}\sum_{y\in\mathbb{Z}^d}\mathrm{tr}\big(|U_y|^2\big) &= \sum_{y\in\mathbb{Z}^d}\sum_{n,m=1}^{s}|\,\langle y,n|\,U\,|0,m\rangle\,|^2 = \sum_{m=1}^{s}\left(\sum_{y\in\mathbb{Z}^d}\sum_{n=1}^{s}|\,\langle y,n|\,U\,|0,m\rangle\,|^2\right) = \sum_{m=1}^{s}\|U\,|0,m\rangle\,\|^2\\ &= \sum_{m=1}^{s}1 = s.\end{aligned}$$

We can therefore interpret $\frac{1}{s}\,\mathrm{tr}\big(|U_y|^2\big)$ as the relative quantity of information moving from 0 to y averaged over the coin space. In this sense, $\mathrm{tr}\big(|U_y|^2\big) = 0$ signifies that independent of the coin state, no information will flow from 0 to y in one application of the unitary operator U. The higher $\mathrm{tr}\big(|U_y|^2\big)$ is, the more information will flow from 0 to y on average.

Hence, $\mathrm{tr}\big(|U_y|^2\big)$ is s times the probability that the walker moves from 0 to y. Therefore the index of U can be viewed as s times the average step that the Quantum Walker will take. It thus corresponds to the average flow of information in a Quantum system governed by U.

In this spirit, we can give a meaning to a 0 index: the index of a Quantum Walk U is 0 if and only if on average there is no flow of information.

Example 4. (Quantum Walks on the hexagonal lattice continued) Using this interpretation, it makes sense that the choice of coin matrix in Example 1 does not influence the index of U. Since we compute the average over the coin space, the effect of the coin operator will be nullified.

Furthermore, we can now easily compute the index of U by considering the shifts used to define the shift operator S. The average time step of a Quantum walker governed by U is given by the sum of the possible shifts divided by 6 (since the coin state is six-dimensional). In Figure 3.1 we see that the shifts cancel each other out, meaning that on average there is no flow of information. This aligns with the zero index we computed in Example 3.

4 Properties of the index

In this chapter we will first characterize the class of matrices that define a Quantum Walk and show that the composition of Quantum Walks corresponds to the multiplication of the respective matrices. This will allow us to prove the additivity of the index. We will then show that $\mathrm{ind}(U)$ is a homotopy invariant and can be computed using the bands of $\hat{U}(k)$.

4.1 Class of matrices which define a Quantum Walk

If we have a closer look at the proof of Proposition 2, we notice that the only requirement for U to reduce to a multiplication by the matrix $\hat{U}(k)$ in Fourier space was actually translation invariance. Locality corresponds to each entry $\left(\hat{U}(k)\right)_{n,m}$ being a Laurent polynomial (in contrast to a Laurent series). As shown in Corollary 1, the unitarity of U corresponds to the unitarity of the matrix $\hat{U}(k)$. Since $\hat{U}(k)$ is unitary for every $k \in \mathbb{T}^d$ and every entry is of the form

$$A(k)_{n,m} = \sum_{j=1}^{J} c_j e^{iy^{(j)}\cdot k} \text{ for all } n, m, \text{ where } y^{(j)} \in \mathbb{Z}^d,\ c_j \in \mathbb{C}, \tag{4.1}$$

it is manifest to define the set

$$\mathcal{M} = \{A(k) \in \mathbb{C}^{s\times s} \mid \text{every entry is of the form (4.1) and } A(k) \text{ is unitary } \forall k \in \mathbb{T}^d\}. \tag{4.2}$$

Corollary 2. *The set $\mathcal{M}$ is a group, its group operation being matrix multiplication.*

Proof. Let $A(k) \in \mathcal{M}$, then its adjoint is again of the form (4.1). Thus, $A(k)^\dagger \in \mathcal{M}$. Furthermore, if $A(k), B(k) \in \mathcal{M}$, then $A(k)B(k)$ is unitary for all k and every entry of $A(k)B(k)$ again a Laurent polynomial in $e^{ik_1}, ..., e^{ik_d}$. Since the identity matrix is an element of $\mathcal{M}$ as well, the set $\mathcal{M}$ together with matrix multiplication forms a group. □

Let $\mathcal{Q}$ denote the group of Quantum Walks on $l^2(\mathbb{Z}^d;\mathbb{C}^s)$. We already know that every Quantum Walk U induces a unique matrix $\hat{U}(k) \in \mathcal{M}$, which allows us to define the map

$$\Psi : \mathcal{Q} \to \mathcal{M},\quad U \mapsto \hat{U}(k).$$

Since $\hat{U}(k)$ uniquely describes the action of U at $0 \in \mathbb{Z}^d$, we conclude, using translation invariance, that U is characterized by $\hat{U}(k)$. Thus, the map Ψ is injective. It turns out that Ψ is in fact a bijection and preserves the group structures. The goal of this section is proving the following:

Theorem 2. *The map $\Psi : \mathcal{Q} \to \mathcal{M}$ is a group isomorphism.*

The proof of Theorem 2 will be split in two parts. We will first show that Ψ is a bijection and then prove that it preserves the group structures.

Lemma 2. *The map $\Psi : \mathcal{Q} \to \mathcal{M}$ is a bijection.*

Proof. It only remains to show the surjectivity of Ψ. For any matrix $A(k) \in \mathcal{M}$, we define the multiplication operator $\hat{U}$ on $L^2(\mathbb{T}^d;\mathbb{C}^s)$:

$$\hat{U} : L^2(\mathbb{T}^d;\mathbb{C}^s) \to L^2(\mathbb{T}^d;\mathbb{C}^s),\ (\hat{U}\hat{\psi})(k) = A(k)\hat{\psi}(k).$$

Let $U = \mathcal{F}^{-1}\hat{U}\mathcal{F}$ be the corresponding operator on $l^2(\mathbb{Z}^d;\mathbb{C}^s)$. Since $A(k)$ is a unitary matrix, the adjoint and inverse of $\hat{U}$ is the multiplication operator by the matrix $A(k)^\dagger$. Hence, $\hat{U}$ is unitary. For any translation T_v on $l^2(\mathbb{Z}^d;\mathbb{C}^s)$ we define $\widehat{T_v} = \mathcal{F}T_v\mathcal{F}^{-1}$ to be the corresponding operator in Fourier space. Its action is given by $\widehat{T_v}\hat{\psi}(k) = e^{-ik\cdot v}\hat{\psi}(k)$. Since both $\widehat{T_v}$ and $\hat{U}$ are multiplication operators, they commute.

Because the Fourier Transform $\mathcal{F}$ is unitary, the operator U is thus unitary and translation invariant. U is local, since the entries of $A(k)$ are Laurent polynomials. Therefore, every matrix $A(k) \in \mathcal{M}$ defines a local, translation invariant, unitary operator U, which means that Ψ is surjective. □

The second and final step in proving Theorem 2 is showing that Ψ preserves the group structures:

Lemma 3. *Let $U, V \in \mathcal{Q}$ be two Quantum Walks. Then we have $\Psi(UV) = \Psi(U)\,\Psi(V)$. In other words, the matrix corresponding to the composition of two Quantum Walks is the matrix product of the two matrices corresponding to the respective Quantum Walks.*

Proof. Let U_1, U_2 be two Quantum Walks and $\hat{U}_1(k), \hat{U}_2(k) \in \mathcal{M}$ their respective matrices. We compute

$$\begin{aligned}\Big(\hat{U}_1(k)\,\hat{U}_2(k)\Big)_{n,m} &= \sum_{l=1}^{s} \hat{U}_1(k)_{n,l}\,\hat{U}_2(k)_{l,m} = \sum_{l=1}^{s}\Big(\sum_{x\in\mathbb{Z}^d}\langle x,n|\,U_1\,|0,l\rangle\; e^{ix\cdot k}\Big)\Big(\sum_{y\in\mathbb{Z}^d}\langle y,l|\,U_2\,|0,m\rangle\; e^{iy\cdot k}\Big)\\ &\overset{\text{fin. sums}}{=} \sum_{l=1}^{s}\sum_{x,y\in\mathbb{Z}^d}\underbrace{\langle x,n|\,U_1\,|0,l\rangle}_{=\langle x,n|U_1\,T_{-y}\,|y,l\rangle}\;\langle y,l|\,U_2\,|0,m\rangle\; e^{i(x+y)\cdot k}\\ &= \sum_{l=1}^{s}\sum_{x,y\in\mathbb{Z}^d}\langle x+y,n|\,U_1\,|y,l\rangle\;\langle y,l|\,U_2\,|0,m\rangle\; e^{i(x+y)\cdot k}.\end{aligned}$$

Here we used that $T_x^\dagger = T_{-x}$ for any $x \in \mathbb{Z}^d$, as shown in chapter 2. We relabel the summation over $x, y \in \mathbb{Z}^d$ above, using the fact that

$$f : \mathbb{Z}^d \times \mathbb{Z}^d \to \mathbb{Z}^d \times \mathbb{Z}^d,\ (x,y) \mapsto (x+y,y)$$

is a bijection. We thus conclude

$$\begin{aligned}\Big(\hat{U}_1(k)\,\hat{U}_2(k)\Big)_{n,m} &= \sum_{l=1}^{s}\sum_{x,y\in\mathbb{Z}^d}\langle x,n|\,U_1\,|y,l\rangle\;\langle y,l|\,U_2\,|0,m\rangle\; e^{ix\cdot k}\\ &= \sum_{x\in Z^d}\langle x,n|\,U_1\Big(\sum_{l=1}^{s}\sum_{y\in\mathbb{Z}^d}|y,l\rangle\,\langle y,l|\Big)U_2\,|0,m\rangle\, e^{ix\cdot k}\\ &= \sum_{x\in Z^d}\langle x,n|\,U_1\,U_2\,|0,m\rangle\; e^{ix\cdot k} = \widehat{U_1U_2}(k)_{n,m}.\end{aligned}$$

This finishes the proof of the Lemma. □

Combining the Lemmas 2 and 3 proves Theorem 2. An easy corollary of this is now the additivity of the index (see for instance [Vog09]):

Corollary 3. *Let U and V be two Quantum Walks, then: $ind(UV) = ind(U) + ind(V)$.*

Proof. We compute

$$\begin{aligned}\det\Big(\widehat{UV}(k)\Big) &= \det\Big(\hat{U}(k)\,\hat{V}(k)\Big) = \det\Big(\hat{U}(k)\Big)\,\det\Big(\hat{V}(k)\Big) = C_1\,e^{i\,\mathrm{ind}(U)\cdot k}\,C_2\,e^{i\,\mathrm{ind}(V)\cdot k}\\ &= C_1C_2e^{i(\mathrm{ind}(U)+\mathrm{ind}(V))\cdot k}.\end{aligned}$$

□

4.2 Homotopy invariance

The set $\mathcal{Q}$ of Quantum Walks is a subset of $\mathcal{B}\big(l^2(\mathbb{Z}^d;\mathbb{C}^s)\big)$, the normed vector space of linear bounded operators on $l^2(\mathbb{Z}^d;\mathbb{C}^s)$. This topology allows us to discuss the notion of homotopy, which in this case means the continuous deformation of operators. We will show the following:

Proposition 3. *Let $\gamma : [0,1] \to \mathcal{B}\big(l^2(\mathbb{Z}^d;\mathbb{C}^s)\big)$, $t \mapsto U_t$ be a norm-continuous map such that the operator U_t is local, translation invariant, and unitary for every $t \in [0,1]$. Then the index is constant along γ.*

Proof. Since U_t is a Quantum Walk, it corresponds to a multiplication operator by the matrix $\hat{U}_t(k)$ in Fourier space, where

$$\big(\hat{U}_t(k)\big)_{n,m} = \sum_{y\in\mathbb{Z}^d} \langle y,n|U_t|0,m\rangle\, e^{iy\cdot k}.$$

Using continuity of the scalar product, the map $(t,k) \mapsto \big(\hat{U}_t(k)\big)_{n,m}$ is continuous. Thus, γ induces a continuous map $\tilde{\gamma} : [0,1] \to \mathcal{M}$, $t \mapsto \hat{U}_t(\cdot)$. Since the determinant is continuous as well, the map

$$(t,k) \mapsto \det\big(\tilde{\gamma}(t)(k)\big) = C_t\, e^{ik\cdot n_t}$$

is continuous. Hence $t \mapsto n_t$ must be continuous as well. Since $n_t \in \mathbb{Z}^d$, the index n_t is constant for all t. Therefore, the index of a Quantum Walk is a homotopy invariant. □

4.3 Relation to the bandstructure

Since $\det\Big(\hat{U}(k)\Big) = C\, e^{ik\cdot v}$, we compute

$$-i\nabla\left(\log\left(\det\left(\hat{U}(k)\right)\right)\right) = -i\nabla\big(\log(C) + ik\cdot v\big) = v = \mathrm{ind}(U).$$

Integrating this over the torus gives

$$\frac{1}{(2\pi)^d\, i}\int_{\mathbb{T}^d} \nabla\left(\log\left(\det\left(\hat{U}(k)\right)\right)\right)\, dk = \frac{1}{(2\pi)^d i}\int_{\mathbb{T}^d} iv\, dk = v = \mathrm{ind}(U). \tag{4.3}$$

Note that in the one-dimensional case $d = 1$, the formula above corresponds to the winding number of the curve $k \in [0,2\pi] \mapsto \det\Big(\hat{U}(k)\Big)$, see for instance [Ced+17; Kit06].
Let $\omega_\alpha(k)$ be the bands of $\hat{U}(k)$, i.e. $e^{i\omega_\alpha(k)}$ are its eigenvalues. We compute

$$\log\Big(\det\Big(\hat{U}(k)\Big)\Big) = \log\left(\prod_{\alpha=1}^{s} e^{i\omega_\alpha(k)}\right) = \log\Big(e^{i\sum_{\alpha=1}^{s}\omega_\alpha(k)}\Big) = i\sum_{\alpha=1}^{s}\omega_\alpha(k).$$

Plugging this into (4.3) leads to

$$\mathrm{ind}(U) = \frac{1}{(2\pi)^d\, i}\int_{\mathbb{T}^d} \nabla\left(i\sum_{\alpha=1}^{s}\omega_\alpha(k)\right) dk = \frac{1}{(2\pi)^d}\sum_{\alpha=1}^{s}\int_{\mathbb{T}^d}\nabla\omega_\alpha(k)\, dk, \tag{4.4}$$

giving us a connection between the index of a Quantum Walk and its bands. Note that so far, we have tacitly assumed the bands to be sufficiently differentiable. In fact, one can show the following:

Lemma 4. *The eigenvalues $e^{i\omega_\alpha(k)}$ of $\hat{U}(k)$ can be parametrized in a way such that the gradients $\nabla\omega_\alpha(k)$ exist almost everywhere.*

The proof of this Lemma relies heavily on [Rai13, Theorem 6.19] and is given in the Appendix A.2. Hence, if the gradients of the bands are well defined at a point $k \in \mathbb{T}^d$, we have additionally (see for instance [Vog09]):

$$\mathrm{ind}(U) = -i\nabla\left(\log\left(\det\left(\hat{U}(k)\right)\right)\right) = \sum_{\alpha=1}^{s} \nabla\omega_\alpha(k). \tag{4.5}$$

Example 5. Let U be the local, translation invariant, unitary operator on $l^2(\mathbb{Z};\mathbb{C}^2)$ such that

$$U : \begin{cases} |0,1\rangle \mapsto \frac{1}{2}\left(|1,1\rangle - |-1,1\rangle + |0,2\rangle + |2,2\rangle\right) \\ |0,2\rangle \mapsto \frac{1}{2}\left(-|1,1\rangle - |-1,1\rangle + |0,2\rangle - |2,2\rangle\right). \end{cases}$$

U defines a Quantum Walk on the line $\mathbb{Z}$ with two-dimensional coin space $\mathbb{C}^2$. Using equation (3.1), its matrix $\hat{U}(k)$ is given by

$$\hat{U}(k) = \frac{1}{2}\begin{pmatrix} e^{ik} - e^{-ik} & -e^{ik} - e^{-ik} \\ 1 + e^{2ik} & 1 - e^{2ik} \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 2i\sin(k) & -2\cos(k) \\ 1 + e^{2ik} & 1 - e^{2ik} \end{pmatrix}.$$

We compute its determinant $\det\left(\hat{U}(k)\right) = e^{ik}$ and conclude $\mathrm{ind}(U) = 1$. We can verify the index by reading off the winding number of $\det\left(\hat{U}(k)\right)$ of the complex curve $k \mapsto k\,\det\left(\hat{U}(k)\right)$:

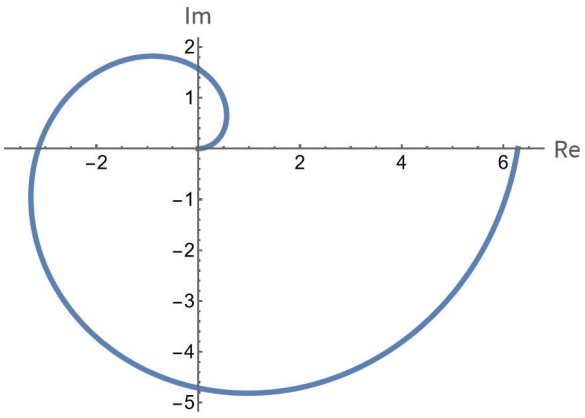
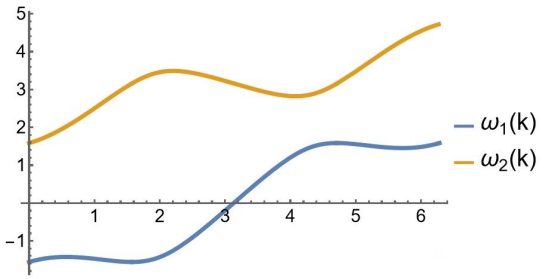
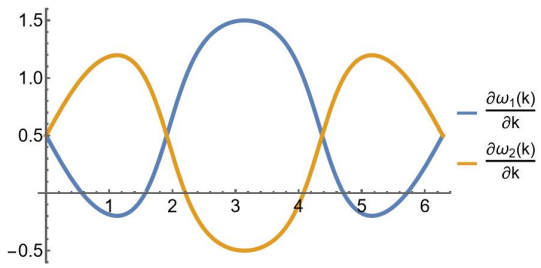


Figure 4.1 The complex curve $k \mapsto k\,\det\left(\hat{U}(k)\right)$

We observe that $\det\left(\hat{U}(k)\right)$ does exactly one counter-clockwise rotation around the origin, which means its winding number is 1. Furthermore, we can compute its bandstructure



Bands of $\hat{U}(k)$



Derivatives of the bands

Figure 4.2 The Bandstructure of $\hat{U}(k)$

It is easy to see that $\omega_2'(k) = 1 - \omega_1'(k)$. Hence, the sum of the derivatives of the bands is 1 for all k, in accordance with (4.5)

5 Link to the index of a pair of projections

In this chapter we will connect the index of a Quantum Walk to the index of a pair of projections. We will show for a Quantum Walk U defined on the one-dimensional grid $\mathbb{Z}$ that its index is given by $\operatorname{tr}\left(U^\dagger PU - P\right)$, where P is the projection onto $\{x \geq 0\}$. This will allow us to apply the theory of the index of a pair of projections developed by Avron, Seiler, and Simon [ASS94]. For Quantum Walks on higher dimensional grids (i.e. $\mathbb{Z}^d$ with $d \geq 2$) we will show a similar relation, even though the required projection P_L will depend on the interaction length L of U.

5.1 One-dimensional case

The main result of this section is the following theorem, which was first shown by Kitaev [Kit06].

Theorem 3. *Let U be a Quantum Walk on $\mathbb{Z}$ and let P be the projection onto the positive integers $\{x \geq 0\}$. Then the operator $U^\dagger PU - P$ is trace class and*

$$ind(U) = \operatorname{tr}\Big(U^\dagger PU - P\Big).$$

Note that P being a projection onto $\{x \geq 0\}$ means that the action of P on an element $|x, n\rangle$ of the orthonormal basis is given by:

$$P\,|x, n\rangle = \begin{cases} |x, n\rangle & \text{if } x \geq 0 \\ 0 & \text{else.} \end{cases}$$

It will turn out that translation invariance is actually not necessary for $\operatorname{tr}\left(U^\dagger PU - P\right)$ to be well-defined and an integer. In this way, we can extend our index to Quantum Walks on $\mathbb{Z}$ which need not be translation invariant.

The proof of Theorem 3 is loosely based on [Kit06] and will be split into two Lemmas. We will first show that $U^\dagger PU - P$ is trace class and then connect it to the index of U. Another proof can for instance be found in [ABJ18], where the requirement of locality is slightly relaxed.

Lemma 5. *Let U be a local, unitary operator on $l^2(\mathbb{Z}; \mathbb{C}^s)$ and P be the projection onto $\{x \geq 0\}$. Then $\Phi = U^\dagger PU - P$ is a finite rank operator, and therefore trace class.*

In order to prove the Lemma above in all its generality, we will need some theory on Quantum Walks that are not translation invariant. Following is the more general definition of a Quantum Walk that is used in [Gro+12] and [Gei22].

We equip every $x \in \mathbb{Z}^d$ with a finite dimensional Hilbert space $\mathcal{H}_x$, assume that the dimensions of $\mathcal{H}_x$ are uniformly bounded, and define the separable Hilbert space

$$\mathcal{H} = \bigoplus_{x \in \mathbb{Z}^d} \mathcal{H}_x.$$

For any operator U on $\mathcal{H}$, let $U_{yx} : \mathcal{H}_x \to \mathcal{H}_y$ be the matrix corresponding to the direct sum above. A Quantum Walk is defined to be a local, unitary operator U on $\mathcal{H}$ (see Chapter 2 for more details). This allows us to prove Lemma 5.

Proof. Let $L \in \mathbb{N}$ be the interaction length of U, i.e. $U_{yx} = 0$ if $|y - x| > L$. This means that

$$U\,|x,m\rangle = \bigoplus_{y\in\mathbb{Z}} U_{yx}\,|x,m\rangle = \bigoplus_{x-L\le y\le x+L} U_{yx}\,|x,m\rangle\,.$$

Consequently, if $x > L$, then $x - L > 0$ and $PU\,|x,m\rangle = U\,|x,m\rangle$. We conclude

$$\Phi\,|x,m\rangle = U^\dagger PU\,|x,m\rangle - P\,|x,m\rangle = U^\dagger U\,|x,m\rangle - |x,m\rangle = |x,m\rangle - |x,m\rangle = 0.$$

On the other hand, if $x < -L$, then $U\,|x,n\rangle = \bigoplus_{y<0} U_{yx}\,|x,n\rangle$, which means that $PU\,|x,n\rangle = 0$. We therefore have

$$\Phi\,|x,n\rangle = U^\dagger PU\,|x,n\rangle - P\,|x,n\rangle = 0.$$

Hence, $\Phi\,|x,n\rangle = 0$ if $|x| > L$. Using self-adjointness, this gives us for $|y| > L$ and any $x \in \mathbb{Z}$:

$$\langle y,n|\,\Phi\,|x,m\rangle = \overline{\langle x,m|\,\Phi\,|y,n\rangle} = 0.$$

Thus, the range of Φ is a subspace of $\operatorname{span}\Big(\{|y,n\rangle \mid y \in \mathbb{Z},\ |y| \le L,\ n = 1,\dots,s\}\Big)$. In other words, Φ is a finite rank operator, which in turn implies Φ being trace class. □

This enables us to connect the index of U as defined in Definition 3 to the trace of $U^\dagger PU - P$:

Lemma 6. *Let U be a local, unitary, translation invariant operator on $l^2(\mathbb{Z};\mathbb{C}^s)$ and P be the projection onto $\{x \in \mathbb{Z} \mid x \ge 0\}$. Then $\mathrm{ind}(U) = \operatorname{tr}\big(U^\dagger PU - P\big)$.*

Proof. Note that $U^\dagger PU - P = U^\dagger[P,U]$. We want to compute the blocks $[P,U]_{yx}$ of the commutator. For this, let $y < 0 \le x$. We compute:

$$\big([P,U]_{yx}\big)_{n,m} = \langle y,n|\ [P,U]\ |x,m\rangle = \langle y,n|\ PU\ |x,m\rangle - \langle y,n|\ UP\ |x,m\rangle = -\langle y,n|\ U\ |x,m\rangle = -\big(U_{yx}\big)_{n,m}\,.$$

Hence, for $x \ge 0 > y$ the block matrix $[P,U]_{yx}$ is given by $-U_{yx}$. Similarly, we compute:

$$[P,U]_{yx} = \begin{cases} 0 & \text{if } x,y \ge 0 \\ U_{yx} & \text{if } y \ge 0 > x \\ -U_{yx} & \text{if } y < 0 \le x \\ 0 & \text{if } x,y < 0 \end{cases}$$

As shown in Lemma 5 the operator $U^\dagger PU - P$ is trace class and its trace is given by:

$$\begin{aligned}
\operatorname{tr}\Big(U^\dagger\,[P,U]\Big) &= \sum_{x\in\mathbb{Z}}\sum_{n=1}^{s}\langle x,n|\ U^\dagger\,[P,U]\ |x,n\rangle = \sum_{x,y\in\mathbb{Z}}\sum_{n=1}^{s}\langle n|\ U_{yx}^\dagger[P,U]_{yx}\ |n\rangle \\
&= \left(\sum_{y\ge 0>x}\sum_{n=1}^{s}\langle n|\ U_{yx}^\dagger U_{yx}\ |n\rangle\right) - \left(\sum_{y<0\le x}\sum_{n=1}^{s}\langle n|\ U_{yx}^\dagger U_{yx}\,|n\rangle\right) \\
&= \sum_{y\ge 0>x}\Big(\operatorname{tr}\big(|U_{yx}|^2\big) - \operatorname{tr}\big(|U_{xy}|^2\big)\Big). \qquad (5.1)
\end{aligned}$$

Since U is assumed to be translation invariant, we can identify U_{yx} and U_{y-x} and connect the trace of $U^\dagger PU - P$ with the index of U, using equation (3.8):

$$\operatorname{tr}\Big(U^\dagger PU - P\Big) = \sum_{y\ge 0>x}\Big(\operatorname{tr}\big(|U_{y-x}|^2\big) - \operatorname{tr}\big(|U_{x-y}|^2\big)\Big) = \sum_{y\in\mathbb{Z}} y\,\operatorname{tr}\big(|U_y|^2\big) = \mathrm{ind}(U). \qquad (5.2)$$

□

In the following we will employ the index theory of a pair of projections developed by Avron, Seiler, and Simon [ASS94]. In the Appendix A.3 we give a small overview of their theory containing some helpful results which we will use in the following lines.

Using Proposition 6 from the Appendix A.3, we know that the pair $(U^\dagger PU, P)$ is a Fredholm pair of projections. Their index is the Fredholm index of the map $PU^\dagger PU$ and given by

$$\operatorname{index}(U^\dagger PU, P) = \dim(\ker(U^\dagger PU - P - 1)) - \dim(\ker(U^\dagger PU - P + 1)) = \operatorname{tr}\Big(U^\dagger PU - P\Big).$$

It is an integer even if U is not assumed to be translation invariant. Furthermore, we remark that the only step in the proof of Lemma 6 where translation invariance was required is equation 5.2, which connects the trace to the index of a Quantum Walk. Therefore, for any local, unitary operator U on $l^2(\mathbb{Z};\mathbb{C}^s)$ we can define its index to be:

$$\operatorname{ind}(U) = \operatorname{tr}\Big(U^\dagger PU - P\Big) = \sum_{y\geq 0>x} \Big(\operatorname{tr}\big(|U_{yx}|^2\big) - \operatorname{tr}\big(|U_{xy}|^2\big)\Big). \tag{5.3}$$

This quantity was first introduced by Kitaev [Kit06], who called it the flow of a local, unitary operator. In fact, it was shown by Cedzich et al. [Ced+17] that this index can be extended to an even broader class of operators, namely essentially local, unitary operators. Essential locality is a generalization of locality, which is then often called strict locality to emphasize the difference. Intuitively, one can think of essentially local operators as being strictly local up to compact perturbations.

We have therefore shown that for Quantum Walks on the one-dimensional grid, their index is in fact a Fredholm index, which now allows us to apply the theory of Fredholm operators. It is immediately clear that the index is a homotopy invariant. Additionally, we can quite easily prove the additivity of the index: Let U_1 and U_2 be two local, unitary operators on $l^2(\mathbb{Z};\mathbb{C}^s)$. Using Proposition 8.1 from the Appendix A.3, we know that

$$\operatorname{index}(U_1^\dagger PU_1\,, P) = \operatorname{index}(U_2^\dagger U_1^\dagger PU_1U_2\,, U_2^\dagger PU_2).$$

The pairs $(U_2^\dagger U_1^\dagger PU_1U_2\,, U_2^\dagger PU_2)$ and $(U_2^\dagger PU_2\,, P)$ are Fredholm pairs of projections. Since $(U_2^\dagger PU_2\,, P)$ is trace-class, it is compact and we can apply Proposition 8.2 to conclude

$$\begin{aligned}\operatorname{ind}(U_1U_2) &= \operatorname{index}(U_2^\dagger U_1^\dagger PU_1U_2\,, P) = \operatorname{index}(U_2^\dagger U_1^\dagger PU_1U_2\,, U_2^\dagger PU_2) + \operatorname{index}(U_2^\dagger PU_2\,, P)\\ &= \operatorname{index}(U_1^\dagger PU_1\,, P) + \operatorname{index}(U_2^\dagger PU_2\,, P) = \operatorname{ind}(U_1) + \operatorname{ind}(U_2).\end{aligned}$$

5.2 Higher dimensions

It would be convenient to be able to find a similar link between the index of a Quantum Walk and the index of a pair of projections on the higher dimensional grid $\mathbb{Z}^d$ as well. Even though there exists no general projection P that leads to such a relation in higher dimensions, we will show that, depending on the interaction length L of the Quantum Walk U, we can give a projection P_L connecting $\operatorname{ind}(U)$ to the index of a pair of projections.

Let U be a Quantum Walk on $\mathbb{Z}^d$, i.e. a local, unitary, translation invariant operator on $l^2(\mathbb{Z}^d;\mathbb{C}^s)$, and let P be a projection onto some set $A \subset \mathbb{Z}^d$. Since $\operatorname{ind}(U)$ is an integer vector and $\operatorname{tr}\big(U^\dagger PU - P\big)$ a scalar, the best we can hope for is finding projections $P_1, ..., P_d$ such that $\operatorname{ind}(U)_j = \operatorname{tr}\big(U^\dagger P_jU - P_j\big)$.

Let $A \subset \mathbb{Z}^d$ and P be the projection onto A. Suppose that the operator $U^\dagger PU - P$ is trace class. We want to establish necessary conditions such that $\operatorname{ind}(U)_1 = \operatorname{tr}\big(U^\dagger PU - P\big)$. Following a similar argument as in the proof of Lemma 6, the blocks of $[P, U]$ are given by:

$$[P,U]_{yx} = \begin{cases} 0 & \text{if } y, x \in A\\ U_{yx} & \text{if } y \in A, x \notin A\\ -U_{yx} & \text{if } y \notin A, x \in A\\ 0 & \text{if } y, x \notin A\end{cases}$$

Assuming that the operator $U^\dagger PU - P$ is trace class, we compute along the lines of (5.1):

$$\begin{aligned}\operatorname{tr}\Big(U^\dagger PU - P\Big) &= \sum_{x,y\in\mathbb{Z}^d}\sum_{n=1}^{s} \langle n|\ (U_{yx})^\dagger\ [P,U]_{yx}\ |n\rangle \\ &= \sum_{y\in A, x\notin A}\sum_{n=1}^{s} \langle n|\, U_{yx}^\dagger U_{yx}\,|n\rangle - \sum_{y\notin A, x\in A}\sum_{n=1}^{s} \langle n|\, U_{yx}^\dagger U_{yx}\,|n\rangle \\ &= \sum_{y\in A,x\notin A} \operatorname{tr}\Big(U_{yx}^\dagger U_{yx}\Big) - \operatorname{tr}\Big(U_{xy}^\dagger U_{xy}\Big) = \sum_{y\in A,x\notin A} \operatorname{tr}\big(|U_{y-x}|^2\big) - \operatorname{tr}\big(|U_{x-y}|^2\big). \end{aligned} \tag{5.4}$$

In the one-dimensional case, using $A = \mathbb{N}_0$, we could connect (5.4) with equation (3.8). To be more precise, the following is true on $\mathbb{Z}$:

$$\sum_{y\geq 0>x} \operatorname{tr}\big(|U_{y-x}|^2\big) - \operatorname{tr}\big(|U_{x-y}|^2\big) = \sum_{y\in\mathbb{Z}} y\ \operatorname{tr}\big(|U_y|^2\big) = \operatorname{ind}(U).$$

This is the crucial step that generally does not hold on $\mathbb{Z}^d$. Thus, the goal is finding a set $A \subset \mathbb{Z}^d$ such that

$$\sum_{y\in A,x\notin A} \operatorname{tr}\big(|U_{y-x}|^2\big) - \operatorname{tr}\big(|U_{x-y}|^2\big) = \sum_{y\in\mathbb{Z}^d} y_1\ \operatorname{tr}\big(|U_y|^2\big). \tag{5.5}$$

Considering the one-dimensional case, our first idea would be to define

$$A = \{y \in \mathbb{Z}^d \mid y_1 \geq 0\}.$$

Note that for every $y^* \in \mathbb{Z}^d$ there exists a Quantum Walk U such that $U_{y^*} \neq 0$ and $U_x = 0$ for all $x \in \mathbb{Z}^d \setminus \{^*y\}$, one instance being the shift operator $|x,n\rangle \mapsto |x+y^*,n\rangle$. Using this with $y^* = e_1$, equation (5.5) implies now:

$$\operatorname{tr}\big(|U_{e_1}|^2\big) = \sum_{y\in A,x\notin A} \underbrace{\operatorname{tr}\big(|U_{y-x}|^2\big) - \operatorname{tr}\big(|U_{x-y}|^2\big)}_{\begin{cases}\operatorname{tr}\left(|U_{e_1}|^2\right) & \text{if } y-e_1 = x\\ -\operatorname{tr}\left(|U_{e_1}|^2\right) & \text{if } y+e_1 = x\\ 0 & \text{else}\end{cases}}. \tag{5.6}$$

However, the case $y \in A, y - e_1 \notin A$ occurs for all y such that $y_1 = 0$, whereas $y \in A, y + e_1 \notin A$ is not possible. We conclude that the right-hand side of (5.6) is not well-defined, showing that (5.5) does not hold when A is defined as above. In fact, one can show the following:

Theorem 4. *Let $d > 1$, then there exists no projection P such that $\operatorname{ind}(U)_1$ is given by (some multiple of) $\operatorname{tr}\big(U^\dagger PU - P\big)$ for all local, translation-invariant, unitary operators U on $l^2(\mathbb{Z}^d;\mathbb{C}^s)$.*

A proof of Theorem 4 is given in the Appendix A.4. Naturally, a similar result holds for all other entries $\operatorname{ind}(U)_j$ of the index. Hence, Definition (5.3) can not be directly generalized to higher dimensions.

However, upon closer inspection of (5.6), the underlying problem appears to be something that we can circumvent: Since the operator U is translation invariant and local, it is already entirely characterized by its action on a finite neighborhood of 0. We therefore gain no new information by computing its trace over $\mathbb{Z}^d$. Thus, we hope that by artificially bounding all coordinates but the first we can treat it like a one-dimensional Quantum Walk and compute the trace as before:

Let U be a local, translation invariant, unitary operator on $l^2(\mathbb{Z}^d;\mathbb{C}^s)$ and let $L \in \mathbb{N}$ be its interaction length. We define the finite set

$$\mathbb{T}_L = {}^{\mathbb{Z}}\!/_{(2L+1)\mathbb{Z}} \simeq \{-L, -L+1, ..., L\}.$$

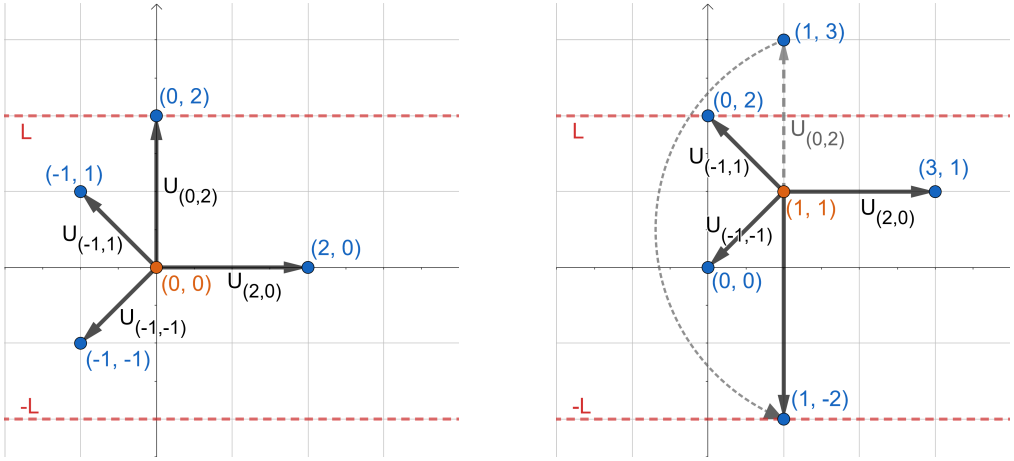
Let $v : \mathbb{Z} \to \{-L, -L+1, ..., L\}$ be the map such that for any $x \in \mathbb{Z}$, $v(x)$ is the unique element of $\{-L, -L+1, ..., L\}$ such that there exists some $n \in \mathbb{Z}$ with $x - (2L+1)n = v(x)$. We then define the map:

$$\phi : \mathbb{Z}^d \to \mathbb{Z} \times \mathbb{T}_L^{d-1}, \; x \mapsto \begin{pmatrix} x_1 \\ v(x_2) \\ \vdots \\ v(x_d) \end{pmatrix}.$$

This allows us to define U_L, the restriction of U onto $l^2(\mathbb{Z} \times \mathbb{T}_L^{d-1}; \mathbb{C}^s)$. The operator is given by its action on elements of the orthonormal basis:

$$\langle y, m | \, U_L \, | x, n \rangle = \langle m | \, U_{\phi(y-x)} \, | n \rangle .$$

The action of U_L is illustrated in Figure 5.1 below:



The action of U at the origin

The action of U_L at $(1, 1)$

Figure 5.1 The construction of U_L

As we see above, the operators U and U_L act the same way in most cases. Whenever U sends some information to a point outside of $\mathbb{Z} \times \mathbb{T}_L^{d-1}$ this point is mapped to its representative inside $\mathbb{Z} \times \mathbb{T}_L^{d-1}$. In Figure 5.1 we see this when information is mapped from $(1, 1)$ to $(1, 3)$ by $U_{(0,2)}$. Since $(1, 3) \notin \mathbb{Z} \times \mathbb{T}_L^{d-1}$, the point is mapped to its representative in $\mathbb{Z} \times \mathbb{T}_L^{d-1}$, which is $(1, -2)$.

By definition, U_L is a translation invariant, local operator with interaction length L. It is still surjective and an isometry, which makes it a unitary operator. Since $l^2(\mathbb{Z} \times \mathbb{T}_L^{d-1}; \mathbb{C}^s)$ and $l^2(\mathbb{Z}; C^s)$ are isomorphic, we can view U_L as a one-dimensional Quantum Walk. This intuition motivates us to proceed as in the one-dimensional case and establish a connection between its index and the index of a pair of projections. We will prove the following:

Theorem 5. *Let U be a local, translation invariant, unitary operator on $l^2(\mathbb{Z}^d; \mathbb{C}^s)$ and U_L its restriction onto $l^2(\mathbb{Z} \times \mathbb{T}_L^{d-1}; \mathbb{C}^s)$ as defined above. Let P_L be the projection onto $A_L = \{x \in \mathbb{Z} \times \mathbb{T}_L^{d-1} \mid x_1 \geq 0\}$, then:*

$$ind(U)_1 = \frac{1}{(2L+1)^{d-1}} \operatorname{tr}\Big(U_L^\dagger P_L U_L - P_L\Big).$$

To prove this, we will proceed as in the one-dimensional case. However, we will omit some of the detailed calculations that were already shown before.

Proof. We define the operator $\Phi_L = U_L^\dagger P_L U_L - P_L$. Let $y \in \mathbb{Z} \times \mathbb{T}_L^{d-1}$ be such that $y_1 > L$. This implies that $P_L U_L \, |y, m\rangle = U_L \, |y, m\rangle$, which gives us

$$\Phi_L \, |y, m\rangle = U_L^\dagger P_L U_L \, |y, m\rangle - P_L \, |y, m\rangle = U_L^\dagger U_L \, |y, m\rangle - |y, m\rangle = |y, m\rangle - |y, m\rangle = 0.$$

Similarly, if $y_1 < -L$, we compute: $\Phi_L |y, m\rangle = 0$. Using self-adjointness of Φ_L, we now have for every $x, y \in \mathbb{Z} \times \mathbb{T}_L^{d-1}$ with $|y_1| > L$:

$$\langle y, m| \Phi_L |x, n\rangle = \overline{\langle x, n| \Phi_L |y, m\rangle} = 0.$$

Hence the range of Φ_L is a subspace of $\text{span}\Big(\{|y, n\rangle \mid y \in \mathbb{Z} \times \mathbb{T}_L^{d-1},\ |y_1| \leq L,\ n = 1, ..., s\}\Big)$. This is a finite dimensional subspace, which means the operator Φ_L is a finite-rank operator and therefore trace class.

By the same computation as in the one-dimensional case, the blocks of $[P_L, U_L]$ are given by

$$[P_L, U_L]_{yx} = \begin{cases} 0 & \text{if } y, x \in A_L \\ U_{\phi(y-x)} & \text{if } y \in A_L, x \notin A_L \\ -U_{\phi(y-x)} & \text{if } y \notin A_L, x \in A_L \\ 0 & \text{if } y, x \notin A_L \end{cases}$$

Repeating the same arguments as in equations (5.1) and (5.4), we can now compute:

$$\text{tr}(\Phi_L) = \sum_{x \in \mathbb{Z} \times \mathbb{T}_N} \sum_{n=1}^{s} \langle x, n| U_L^\dagger [P_L, U_L] |x, n\rangle = ... = \sum_{y \in A_L, x \notin A_L} \text{tr}\big(|U_{\phi(y-x)}|^2\big) - \text{tr}\big(|U_{\phi(x-y)}|^2\big).$$

However, we note that for every $z \in \mathbb{Z} \times \mathbb{T}_L^{d-1}$ with $z_1 \geq 0$ there are only finitely many pairs $y \in A_L, x \notin A_L$ such that $z = \phi(y - x)$. The number of pairs depends only on the first coordinate of z. To be precise, there are exactly $z_1(2L+1)^{d-1}$ such pairs. Similarly, for every $z \in \mathbb{Z} \times \mathbb{T}_L^{d-1}$ with $z_1 < 0$, there are $-z_1(2L+1)^{d-1}$ pairs $y \in A_L, x \notin A_L$ such that $z = x - y$. Hence

$$\text{tr}(\Phi_L) = (2L+1)^{d-1} \sum_{y \in \mathbb{Z} \times \mathbb{T}_N^{d-1}} y_1 \ \text{tr}\big(|U_y|^2\big).$$

Since $U_y = 0$ for all $y \notin \mathbb{Z} \times \mathbb{T}_N^{d-1}$, this finally gives us

$$\text{ind}(U)_1 = \sum_{y \in \mathbb{Z}^d} y_1 \ \text{tr}\big(|U_y|^2\big) = \frac{1}{(2L+1)^{d-1}} \ \text{tr}(\Phi_L).$$

□

Remark 3. *For an arbitrary local, unitary operator U on $l^2(\mathbb{Z}^d; \mathbb{C}^s)$ it is not immediately clear whether an index can be defined using the index of a pair of projections. Nevertheless, its existence can be a helpful tool, as shown for instance in [ABJ18; ABJ19; AM19]. Here, the authors prove under the assumption of a non-zero index the existence of a wandering subspace and certain spectral properties, such as the absolutely continuous spectrum covering the whole unit circle. They illustrate these results on the Chalker-Coddington model and Coined Quantum Walks.*

6 Summary and Outlook

We started by showing that the Fourier Transform $\hat{U} = \mathcal{F}U\mathcal{F}^{-1}$ of any local, translation invariant, unitary operator U on the Hilbert space $l^2(\mathbb{Z}^d;\mathbb{C}^s)$ is a multiplication operator by a matrix $\hat{U}(k)$, whose entries are Laurent polynomials. Its determinant is a monomial $Ce^{iv\cdot k}$, which allowed us to define the index of U to be the unique exponent v of the monomial. The index corresponds to the average flow of information, where the average is taken over the coin space. We then showed that it is an additive homotopy invariant and can be computed as the sum of the gradients of the bands ω_α. Furthermore, if $d = 1$, it corresponds to the winding number of $k \mapsto \det\Big(\hat{U}(k)\Big)$. Moreover, the index of a one-dimensional Quantum Walk U is given by $\operatorname{tr}\big(U^\dagger PU - P\big)$, where P is the projection onto $\{x \geq 0\}$. This quantity is the index of the pair of projections $(U^\dagger PU, P)$, which enabled us to use the existing theory developed by [ASS94] to define the index for any local, unitary operator on $l^2(\mathbb{Z};\mathbb{C}^s)$. Additionally, this gave us a simple proof of the homotopy invariance and the additivity of $\operatorname{ind}(U)$. It is a more involved question whether a similar relation holds in higher dimensions. For a given operator U and its interaction length L, we restricted all directions of the d-dimensional grid $\mathbb{Z}^d$ but the first to the bounded set $\{-L, ..., L\}$ and, similar to the one-dimensional case, defined the projection P_L onto the set $\{x_1 \geq 0\}$. The index of U is then given by $\frac{1}{(2L+1)^{d-1}} \operatorname{tr}\Big(U_L^\dagger P_L U_L - P_L\Big)$, where U_L is the restriction of U. However, this construction requires translation invariance and does not directly generalize to any local, unitary operator.

We already mentioned in Remark 3 that the existence of the index for an operator on the higher dimensional grid $\mathbb{Z}^d$ can be a helpful tool that, for instance, may allow us to show certain spectral properties. It is therefore an intriguing question whether a bigger (or maximal) class of operators exists for which such an index can be defined. As we saw in section 5.2, this will most likely involve restricting $d-1$ directions of Z^d to some bounded set. Let $N \in \mathbb{N}$ and define the set $\Lambda_N = \mathbb{Z} \times \{-N, ..., N\}^{d-1}$. For any local, unitary operator U on $l^2(\mathbb{Z}^d;\mathbb{C}^s)$ we define U_N to be its restriction to the set Λ_N, using a similar construction as in Figure 5.1, and the sequence

$$a_N = \frac{1}{(2N+1)^{d-1}} \operatorname{tr}\Big(U_N^\dagger P_N U_N - P_N\Big),$$

where P_N is again the projection onto $\{x \in \Lambda_N \mid x_1 \geq 0\}$. We know from Theorem 5 that if U is assumed to be translation invariant, then the sequence a_N is constantly equal to $\operatorname{ind}(U)_1$ for all $N \geq L$. Thus, as a first step, we can hope to find some class of operators for which the sequence a_N converges. A second goal would be to find criteria for the limit to be contained in some discrete set.

For a given local, unitary operator, we can show that the difference $a_{N+1} - a_N$ is of order $\frac{1}{N}$. However, we have no information about the monotonicity of the sequence and thus can not conclude its convergence. Nevertheless, for the self-adjoint operator $\Phi = U^\dagger PU - P$, the sequence a_N is given by

$$a_N = \frac{1}{(2N+1)^{d-1}} \operatorname{tr}\big(\mathbb{1}_{\Lambda_N}\Phi\big), \tag{6.1}$$

where $\mathbb{1}_{\Lambda_N}$ is the projection onto Λ_N. Here one can notice similarities to the theory of ergodic operators. To be more precise, the following is implied by Birkhoff's theorem (see [AW15, Theorem 3.14]):

Proposition 4. *Let $\Phi(\omega)$ be a standard ergodic operator on $l^2(\mathbb{Z}^d)$ and $f \in C_0(\mathbb{R})$. Then for almost all ω the limit*

$$\lim_{N\to\infty} \frac{1}{(2N+1)^d} \operatorname{tr}\big(\mathbb{1}_{\mathbb{T}_N} f(\Phi(\omega))\big)$$

exists and its value, which does not depend on ω, can be computed using the average of the local spectral measure. Here, $\mathbb{T}_N = \{-N, ..., N\}^d$.

When speaking of a standard ergodic operator $\Phi(\omega)$ on $l^2(\mathbb{Z}^d)$, we can think of the parameter ω to be an element of the probability space $\mathbb{R}^{\mathbb{Z}^d}$ together with the product σ-algebra of Borel sets. The probability measure $\mathbb{P}$ is such that the variables $(\omega(\xi))_{\xi\in\mathbb{Z}^d}$ are independent and identically distributed random variables. We now require $\Phi(\omega)$ to satisfy

$$\Phi(T_x\omega) = T_x\Phi(\omega)T_x^\dagger.$$

Here, we define the translation T_x on elements of the probability space by $T_x\omega(\xi) = \omega(\xi - x)$. For a thorough review of ergodic operators and their theory, we refer to [AW15]. Note that in (6.1) the set Λ_N grows only in $d-1$ directions when $N \to \infty$. Even though $\Phi\,|x, n\rangle = 0$ if $|x_1| > L$, we can not directly apply Proposition 4. For this, we need to express (6.1) in terms of measure-preserving, ergodic shifts on the $d-1$-dimensional grid $\mathbb{Z}^{d-1}$. With this, we could apply Birkhoff's Theorem to conclude the convergence of the sequence a_N for any random operator $U(\omega)$ such that $\Phi(\omega)$ is a standard ergodic operator and therefore extend the index to this class of operators. Nevertheless, it would still remain unclear whether the limit lies in some discrete set.

A Appendix

A.1 The zeros of a multivariate polynomial

Following is a result about the zeros of a multivariate polynomial, which was needed in the proof of Theorem 1.

Corollary 4. *Let $\mathbb{K}$ be a field and $S_1, ..., S_d$ be infinite subsets of $\mathbb{K}$. Let $f(X_1, ..., X_d)$ be a polynomial in d variables over $\mathbb{K}$. If $f(z_1, ..., z_d) = 0$ for all $z_i \in S_i$, then $f = 0$.*

A proof of Corollary 4 can for instance be found in S. Lang's book on Algebra [Lan02, p. 176].

A.2 Differentiability of the bands

Following is the proof of Lemma 4. It relies on Theorem 6.19 in [Rai13]. A simplified version of this theorem is:

Theorem 6. *Let $x \mapsto (A_{n,m}(x))_{1 \leq n,m \leq s}$ be a parametrized family of complex, normal matrices defined on an open set $V \subset \mathbb{R}^d$. If every entry $x \mapsto A_{n,m}(x)$ is C_Q on V, then for any compact set $K \subset V$ there exists a relatively compact neighborhood W of K (i.e. $K \subset W \subset V$) and a parametrization λ_α of the eigenvalues of A on W such that the classical gradient $\nabla \lambda_\alpha(x)$ exists for almost all $x \in W$.*

Here, C_Q denotes the class of functions $f : V \to \mathbb{C}$ such that both $\mathrm{Re}(f)$ and $\mathrm{Im}(f)$ are quasi-analytic. Note that other definitions of C_Q are possible, in general, C_Q can be assumed to be a subalgebra of $C^\infty(U)$ satisfying certain conditions, see [Rai13, p. 5]. With this we can prove Lemma 4:

Proof. Let $V = (0, 2\pi)^d$ and $\hat{U}$ be the matrix valued function $\hat{U} : V \to \mathbb{C}^{s \times s}$, $k \mapsto \hat{U}(k)$. Since $\hat{U}(k)$ is unitary, it is a normal matrix. Every entry is of the form

$$\hat{U}_{n,m}(k) = \sum_{y \in \mathbb{Z}^d} \langle y, n| \, U \, |0, m\rangle \; e^{iy \cdot k} = \sum_{y \in \mathbb{Z}^d} \langle y, n| \, U \, |0, m\rangle \; (\cos(y \cdot k) + i \sin(y \cdot k)) \, .$$

Hence, its real and imaginary parts are real analytic functions, and therefore in C_Q. For every $n \in \mathbb{N}$ we define the set $K_n = [\frac{1}{n}, 2\pi - \frac{1}{n}]^d \subset V$. Then, by the theorem above, there exists a parametrization of the eigenvalues $\lambda_\alpha(k)$ of $\hat{U}(k)$ such that $\lambda_\alpha(k)$ is differentiable for almost all $k \in K_n$. Without loss of generality, we can assume that for $m \leq n$ the parametrization on K_m is the restriction of the parametrization on K_n. We thus get a parametrization of the eigenvalues λ_α on $(0, 2\pi)^d$ that is differentiable almost everywhere.

To get the bands ω_α from the eigenvalues λ_α we need to apply the complex logarithm. However, we must be careful which branch of the complex logarithm we choose, since it has no holomorphic extension onto all of $\mathbb{C}$. Nevertheless, it suffices to pick a branch cut $U \subset \mathbb{C}$ such that $\lambda_\alpha(k) \in U$ for almost all $k \in (0, 2\pi)^d$ and all $\alpha = 1, ..., s$. In the following lines, we will argue why this is always possible:

We fix $\alpha = 1, ..., s$ and let μ denote the Lebesgue measure on $(0, 2\pi)^d$. For all $\theta \in [0, 2\pi)$ we define the set $U_\theta = \lambda_\alpha^{-1}(e^{i\theta})$. For $n \in \mathbb{N}$ let $M_n = \{\theta \in [0, 2\pi) \mid \mu(U_\theta) \geq \frac{1}{n}\}$. Since the sets $(U_\theta)_{\theta \in [0,2\pi)}$ are disjoint, we have

$$\mu\left((0, 2\pi)^d\right) \geq \mu\Big(\bigcup_{\theta \in [0,2\pi)} U_\theta\Big) \geq \mu\Big(\bigcup_{\theta \in M_n} U_\theta\Big) \geq \frac{1}{n} |M_n|.$$

Thus, the sets M_n must be finite. Therefore the set $\{\theta \in [0, 2\pi) \mid \mu(U_\theta) > 0\} = \bigcup_{n\in\mathbb{N}} M_n$ is countable as the countable union of finite sets. Repeating this process for all α implies that there are only countably many $\theta \in [0, 2\pi)$ such that there exists an α such that $\lambda_\alpha^{-1}(e^{i\theta})$ has positive measure.

We can therefore choose a branch cut $U \subset \mathbb{C}$ for the complex logarithm such that $\lambda_\alpha(k) \in U$ for almost all $k \in (0, 2\pi)^d$ and all α. Since log is holomorphic on U, the bands $\omega_\alpha(k) = -i \log(\lambda_\alpha(k))$ are differentiable at almost all $k \in (0, 2\pi)^d$. □

A.3 The index of a pair of projections

This section aims to state some useful results of the index theory of a pair of projections. All results are based on the work of Avron, Seiler, and Simon [ASS94].

Definition 4. *Let Q, P be two orthogonal projections on a separable Hilbert space. The pair (Q, P) is called Fredholm if the map $C = PQ$ is a Fredholm operator when regarded as a map between $Ran(Q)$ and $Ran(P)$. The index of the pair (Q, P) is now defined to be the Fredholm index of C.*

The authors then state two necessary and sufficient conditions for the pair (Q, P) to be a Fredholm pair.

Proposition 5. *The pair (Q, P) is Fredholm if and only if the kernels $\ker(Q - P \pm 1)$ are finite dimensional. In that case it holds that*

$$index(Q, P) = \dim(\ker(Q - P - 1)) - \dim(\ker(Q - P + 1)).$$

Proposition 6. *The pair (Q, P) is a Fredholm pair if and only if there exist two self-adjoint operators D, F such that $\|D\| < 1$, F is a finite rank operator and $Q - P = D + F$.*

It was already shown by Effros [EG89] that $\mathrm{tr}(Q - P)$ is an integer if $Q - P$ is trace class. Avron, Seiler, and Simon go one step further and show the following:

Proposition 7. *Let Q, P be self-adjoint projections such that $Q - P$ is in the trace ideal $\mathcal{J}_{2n+1}$ for some $n \in \mathbb{N}$. Then*

$$\mathrm{tr}\big((Q - P)^{2n+1}\big) = index(Q, P) = \mathrm{tr}\big((Q - P)^{2m+1}\big) \quad \forall\, m \geq n$$

Furthermore, the authors prove the following:

Proposition 8. *Let (Q, P) and (P, R) be Fredholm pairs of projections and U a unitary operator, then*

1. *$(U^\dagger QU, U^\dagger PU)$ is a Fredholm pair and $index(U^\dagger QU, U^\dagger PU) = index(Q, P)$*
2. *If $Q - P$ or $P - R$ are compact, then (Q, R) is a Fredholm pair and*

$$index(Q, R) = index(Q, P) + index(P, R)$$

A.4 Proof of Theorem 4

The aim of this section is the proof of Theorem 4. In other words, we will show that for $d \geq 2$, there exists no set $A \subset \mathbb{Z}^d$ satisfying:

$$\sum_{y\in A, x\notin A} \mathrm{tr}\big(|U_{y-x}|^2\big) - \mathrm{tr}\big(|U_{x-y}|^2\big) = \sum_{y\in\mathbb{Z}^d} y_1 \ \mathrm{tr}\big(|U_y|^2\big). \tag{A.1}$$

The proof of this will be split in two Lemmas. The first Lemma will specify the conditions that the set A needs to fulfill for (A.1) to hold. In the second Lemma, we will show that no set A satisfying these conditions can exist for $d \geq 2$.

Lemma 7. *Let $A \subset \mathbb{Z}^d$ and assume that it satisfies* (A.1) *for all Quantum Walks U. Then, for every $v \in \mathbb{Z}^d$, the set $(A+v) \cap A^C$ must be finite, and both A and its complement A^C must contain infinitely many elements.*

Proof. Suppose $A \subset \mathbb{Z}^d$ is such that (A.1) holds. For any $v \in \mathbb{Z}^d$ we define the sets

$$B_v = \{y \in A \mid y - v \notin A\} = A \cap (A^C + v)$$
$$C_v = \{y \in A \mid y + v \notin A\} = A \cap (A^C - v).$$

Similar to what we did in (5.6), we conclude from (A.1) that for any $v \in \mathbb{Z}^d$:

$$v_1 = |B_v| - |C_v|. \tag{A.2}$$

Thus, the sets B_v and C_v must be finite, otherwise (A.2) would not be well-defined. Note that $|C_v| = |(A+v) \cap A^C|$, which shows the first part of the Lemma.

Furthermore, if A were finite, then for every $y \in A$ such that $y - v \in A^C$, there would be a $y' \in A$ such that $y' + v \in A^C$ and vice versa. Thus, if A were finite, the difference $|B_v| - |C_v|$ would always be 0. The same holds for A^C. Hence, both A and A^C must contain infinitely many elements. □

We remark that in the one-dimensional case we used the set $A = \mathbb{N}_0$, which satisfies these conditions. In the following, we will show that this, however, is not possible in higher dimensions.

Lemma 8. *Let $d > 1$ and $A \subset \mathbb{Z}^d$ be such that both A and its complement A^C contain infinitely many elements. Then there must be some vector $v \in \mathbb{Z}^d$ such that the set $(A+v) \cap A^C$ contains infinitely many elements.*

Proof. We will show the statement only for $d = 2$. We note, however, that all arguments work in higher dimensions as well. We will prove the lemma by contradiction, i.e. we suppose that

$$\forall\, v \in \mathbb{Z}^2 :\ (A+v) \cap A^C \ \text{ contains only finitely many elements.} \tag{A.3}$$

For any integer $m \in \mathbb{Z}$ we define the horizontal line

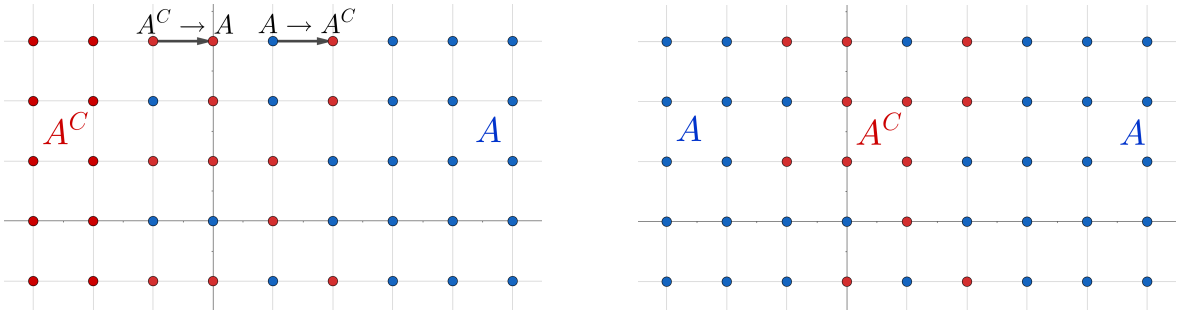
$$H_m = \{x \in \mathbb{Z}^2 \mid x_2 = m\}.$$

In the following, when talking about a transition $A \to A^C$ on H_m, we mean a point $x \in H_m$ such that $x \in A$ and $x + e_1 \in A^C$. Similarly, we define a transition $A^C \to A$, see Figure A.1.

Assumption (A.3) implies that there can only be finitely many transitions $A \to A^C$ and $A^C \to A$ on each line H_m, since otherwise the sets $(A+e_1) \cap A^C$ or $(A-e_1) \cap A^C$ would not be finite. This means that there exist constants $c_m, d_m \in \mathbb{Z}$ such that the intervals $(-\infty, c_m] \times \{m\}$ and $[d_m, \infty) \times \{m\}$ are contained in either A or its complement A^C. Without loss of generality we assume $[d_0, \infty) \times \{0\} \subset A$ (otherwise we just relabel A and A^C).

It follows that $[d_m, \infty) \times \{m\}$ is contained in A for all other $m \in \mathbb{Z}$ as well, since otherwise there would be a vector $v \in \mathbb{Z}^2$ such that $\big([d_0, \infty) \times \{0\}\big) + v \subset A^C$, contradicting the assumption (A.3).

By the same reasoning, the left intervals $(-\infty, c_m] \times \{m\}$ are either all contained in A or all contained in A^C. If they are all contained in A^C (see the first case in Figure A.1), this means that in every horizontal line H_m there is at least one transition $A^C \to A$, which in turn implies that $(A - e_1) \cap A^C$ is not finite. On the other hand, if all left intervals are contained in A (see the second case in Figure A.1), this means that the set A^C is fully contained in the "central strip" made up from the intervals $[c_m, d_m] \times \{m\}$. Since A^C is, however, required to contain infinitely many elements, this, in turn, implies that there must be infinitely many horizontal lines H_m containing an element of A^C. As above, this entails that the set $(A - e_1) \cap A^C$ can not be finite.



The first case The second case

Figure A.1 Transitions $A \to A^C$, $A^C \to A$ and the two different cases

Thus, assumption (A.3) leads to a contradiction in all possible cases. Therefore there must be some translation $v \in \mathbb{Z}^2$ such that $(A + v) \cap A^C$ is infinite. □

List of Figures

3.1 The shifts on the hexagonal lattice 7

4.1 The complex curve $k \mapsto k \det\left(\hat{U}(k)\right)$ 17

4.2 The Bandstructure of $\hat{U}(k)$ 17

5.1 The construction of U_L 22

A.1 Transitions $A \to A^C$, $A^C \to A$ and the two different cases 29

Bibliography

[AW15] M. Aizenman and S. Warzel. *Random Operators: Disorder Effects on Quantum Spectra and Dynamics*. Vol. 168. Graduate Studies in Mathematics. AMS, 2015. ISBN: 978-1-4704-1913-4.

[Amb04] A. Ambainis. *Quantum walks and their algorithmic applications*. 2004. arXiv: quant-ph/0403120 [quant-ph].

[ABJ18] J. Asch, O. Bourget, and A. Joye. *Chirality induced Interface Currents in the Chalker Coddington Model*. 2018. arXiv: 1708.02120 [math-ph].

[ABJ19] J. Asch, O. Bourget, and A. Joye. *Engineering stable quantum currents at bulk boundaries*. June 2019. arXiv: 1906.08181 [math-ph].

[AM19] J. Asch and M. Mouneime. *Examples for stable quantum currents*. Aug. 2019. arXiv: 1908.11689 [math-ph].

[ASS94] J. Avron, R. Seiler, and B. Simon. "The Index of a Pair of Projections". In: *Journal of Functional Analysis* 120.1 (1994), pp. 220–237. ISSN: 0022-1236.

[CGW22] C. Cedzich, T. Geib, and R. F. Werner. "An algorithm to factorize quantum walks into shift and coin operations". In: *Letters in Mathematical Physics* 112.5 (2022).

[Ced+18] C. Cedzich et al. "Complete homotopy invariants for translation invariant symmetric quantum walks on a chain". In: *Quantum* 2 (2018), p. 95.

[Ced+17] C. Cedzich et al. "The Topological Classification of One-Dimensional Symmetric Quantum Walks". In: *Annales Henri Poincaré* 19.2 (Nov. 2017), pp. 325–383.

[EG89] E. G. Effros and A. Grothendieck. "Why the circle is connected: An introduction to quantized topology". In: *The Mathematical Intelligencer* 11 (1989), pp. 27–35.

[Gei22] T. Geib. "Topological Classification of Symmetric Quantum Walks". PhD thesis. Mar. 2022.

[Gro+12] D. Gross, V. Nesme, H. Vogts, and R. Werner. "Index Theory of One Dimensional Quantum Walks and Cellular Automata". In: *Communications in Mathematical Physics* 310 (Mar. 2012), pp. 419–454.

[Gro96] L. K. Grover. *A fast quantum mechanical algorithm for database search*. 1996. arXiv: quant-ph/9605043 [quant-ph].

[KV90] G. Karlsson and M. Vetterli. "Theory of two-dimensional multirate filter banks". In: *Acoustics, Speech and Signal Processing, IEEE Transactions on* 38 (July 1990), pp. 925 –937.

[Kit06] A. Kitaev. "Anyons in an exactly solved model and beyond". In: *Annals of Physics* 321.1 (Jan. 2006), pp. 2–111.

[Lan02] S. Lang. *Algebra*. New York, NY: Springer, 2002. ISBN: 9781461300410 146130041X.

[MN99] J. R. Magnus and H. Neudecker. *Matrix Differential Calculus with Applications in Statistics and Econometrics*. Second. John Wiley, 1999. ISBN: 9781119541202.

[MLK77] M. Morf, B. Levy, and S.-Y. Kung. "New results in 2-D systems theory, part I: 2-D polynomial matrices, factorization, and coprimeness". In: *Proceedings of the IEEE* 65.6 (1977), pp. 861–872.

[Rai13] A. Rainer. "Perturbation theory for normal operators". In: *Transactions of the American Mathematical Society* 365.10 (Feb. 2013), pp. 5545–5577.

[SC19] A. Sarkar and C. M. Chandrashekar. "Multi-bit quantum random number generation from a single qubit quantum walk". In: *Scientific Reports* 9.1 (Aug. 2019).

[Som23] SomeCallMeTim. *Characterizing a complex multivariate polynomial by its zeroset*. Mathematics Stack Exchange. May 2023. eprint: https://math.stackexchange.com/q/4692446.

[Sta18] C. Stahl. "Translation Invariant QuantumWalks with Discrete Symmetries". PhD thesis. 2018.

[Vog09] H. Vogts. "Discrete Time Quantum Lattice Systems". PhD thesis. June 2009.